Algebra I: Class Notes

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April 22, 2026

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1 Lecture 1: Group Actions & The Orbit-Stabilizer Theorem (Jan 12)

1.1 Permutations and Automorphisms

Definition 1.1 (Symmetric Group)**.** *Let X be a set. The set of all permutations (bijections) of X is denoted by Sym_X (or sometimes Σ_X). Under function composition, Sym_X forms a group.*

Definition 1.2 (Automorphism Group)**.** *Let $(G, \cdot)$ be a group. An automorphism of G is a bijection $\phi : G \to G$ that is also a homomorphism, i.e.,*

$$\phi(g_1 g_2) = \phi(g_1)\phi(g_2) \quad \forall g_1, g_2 \in G.$$

The set of all such automorphisms is denoted by $\mathrm{Aut}(G)$. It forms a group under composition.

1.2 Group Actions

Definition 1.3 (Group Action)**.** *Let G be a group and X be a set. An **action** of G on X is a homomorphism $\phi : G \to \mathrm{Sym}_X$.*
We typically write the action as $g \cdot x := \phi(g)(x)$. This notation satisfies two axioms (equivalent to the homomorphism property):

1. ***Identity:*** *$1 \cdot x = x$ for all $x \in X$.*
2. ***Compatibility:*** *$g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x$ for all $g_1, g_2 \in G, x \in X$.*

Theorem 1.4 (Equivalence Relation on X)**.** *Let G act on X. Define a relation $\sim$ on X by:*

$$x_1 \sim x_2 \iff \exists g \in G \text{ such that } g \cdot x_1 = x_2.$$

Then $\sim$ is an equivalence relation.

Proof. We check the properties:

1. **Reflexive:** $1 \cdot x = x \implies x \sim x$.
2. **Symmetric:** If $g \cdot x_1 = x_2$, acting by g^{-1} gives $g^{-1} \cdot (g \cdot x_1) = g^{-1} \cdot x_2 \implies (g^{-1}g) \cdot x_1 = g^{-1} \cdot x_2 \implies x_1 = g^{-1} \cdot x_2$. Thus $x_2 \sim x_1$.
3. **Transitive:** If $g \cdot x_1 = x_2$ and $h \cdot x_2 = x_3$, then $h \cdot (g \cdot x_1) = x_3 \implies (hg) \cdot x_1 = x_3$. Thus $x_1 \sim x_3$.

□

1.3 Orbits and Stabilizers

The equivalence classes under the relation $\sim$ partition the set X. These classes are called **orbits**.

Definition 1.5 (Orbit)**.** *For $x \in X$, the orbit of x is the set of all places x can be moved to by G:*

$$\mathcal{O}_x = Orb(x) = \{g \cdot x : g \in G\}.$$

Definition 1.6 (Stabilizer)**.** *For $x \in X$, the stabilizer of x is the set of elements in G that fix x:*

$$G_x = \mathrm{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

Note that G_x is a subgroup of G (denoted $G_x \leq G$).

Theorem 1.7 (**Orbit-Stabilizer Theorem**)**.** *Let G act on X. For any $x \in X$, there is a bijection between the orbit $\mathcal{O}_x$ and the set of left cosets G/G_x. Consequently:*

$$|\mathcal{O}_x| = [G : G_x].$$

If G is finite, $|G| = |\mathcal{O}_x| \cdot |G_x|$.

Proof. Define a map $\psi : G/G_x \to \mathcal{O}_x$ by $\psi(gG_x) = g \cdot x$.

1. **Well-defined:** Suppose $g_1G_x = g_2G_x$. Then $g_1^{-1}g_2 \in G_x$, so $(g_1^{-1}g_2) \cdot x = x$, implying $g_2 \cdot x = g_1 \cdot x$.
2. **Injectivity:** If $\psi(g_1G_x) = \psi(g_2G_x)$, then $g_1 \cdot x = g_2 \cdot x$. Multiplying by g_1^{-1}, we get $x = g_1^{-1}g_2 \cdot x$, so $g_1^{-1}g_2 \in G_x$, implying $g_1G_x = g_2G_x$.
3. **Surjectivity:** By definition, any $y \in \mathcal{O}_x$ is of the form $g \cdot x$ for some g, which is exactly $\psi(gG_x)$.

Thus, ψ is a bijection. $\square$

2 Lecture 2: Cauchy's Theorem & Conjugation (Jan 14)

2.1 Cauchy's Theorem

Lemma 2.1. *If X is a finite set and G is a p-group (a group of order p^k) acting on X, then:*

$$|X| \equiv |X^G| \pmod{p},$$

where $X^G = \{x \in X : g \cdot x = x, \forall g \in G\}$ is the set of fixed points.

Theorem 2.2 (Cauchy's Theorem)**.** *If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.*

Proof (McKay's Proof). Let X be the set of p-tuples of elements of G whose product is the identity:

$$X = \{(g_1, \ldots, g_p) \in G^p : g_1 g_2 \cdots g_p = 1\}.$$

Notice that g_p is uniquely determined by the first $p-1$ elements via $g_p = (g_1 \cdots g_{p-1})^{-1}$, so $|X| = |G|^{p-1}$. Since $p \mid |G|$ and $p - 1 \geq 1$, we have $p \mid |X|$.
Let $\mathbb{Z}_p$ (cyclic group of order p) act on X by cyclic shift:

$$\sigma \cdot (g_1, g_2, \ldots, g_p) = (g_2, g_3, \ldots, g_p, g_1).$$

This is a valid action because if $g_1 \cdots g_p = 1$, then $g_2 \cdots g_p = g_1^{-1}$, so $(g_2 \cdots g_p)g_1 = g_1^{-1} g_1 = 1$, which shows the shifted tuple is still in X.
By the Lemma, $|X| \equiv |X^{\mathbb{Z}_p}| \pmod{p}$. Fixed points are exactly tuples $(a, a, \ldots, a)$ such that $a^p = 1$. Since $(1, 1, \ldots, 1) \in X^{\mathbb{Z}_p}$, the set of fixed points is non-empty. Since $p \mid |X|$ and $|X| \equiv |X^{\mathbb{Z}_p}| \pmod{p}$, we have $p \mid |X^{\mathbb{Z}_p}|$. Therefore $|X^{\mathbb{Z}_p}| \geq p$, so there must exist some $a \neq 1$ with $a^p = 1$; this element has order exactly p. □

2.2 Conjugation

Conjugation is a specific action of G on itself.

Definition 2.3. *For $g, h \in G$, the **conjugate** of g by h is $g^h := hgh^{-1}$. The map $\phi_h : G \to G$ defined by $g \mapsto hgh^{-1}$ is an automorphism (an inner automorphism).*

Definition 2.4 (Classes and Centralizers)**.** *Let G act on itself by conjugation ($h \cdot g = hgh^{-1}$).*

- *The orbit of g is the **Conjugacy Class** of g: $Cl(g) = \{hgh^{-1} : h \in G\}$.*
- *The stabilizer of g is the **Centralizer** of g: $C_G(g) = \{h \in G : hg = gh\}$.*

By Orbit-Stabilizer: $|Cl(g)| = [G : C_G(g)]$.

Definition 2.5 (Center of G)**.** *The* ***Center****, $Z(G)$, is the set of elements that commute with everything:*

$$Z(G) = \{z \in G : zg = gz, \forall g \in G\} = \bigcap_{g \in G} C_G(g).$$

$Z(G)$ is the kernel of the conjugation homomorphism $G \to \mathrm{Aut}(G)$. Thus $Z(G) \trianglelefteq G$.

Definition 2.6 (Normalizer)**.** *Let $H \leq G$. The* ***Normalizer*** *of H in G is:*

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

$N_G(H)$ is the largest subgroup of G in which H is normal.

3 Lecture 3: Sylow Preliminaries & Isomorphism Theorems (Jan 16)

3.1 Product of Subgroups

Definition 3.1. *Let $K, N \leq G$. Define $KN = \{kn : k \in K, n \in N\}$.*

- *In general, KN is not a subgroup.*
- *If $N \trianglelefteq G$ (or just $N \subseteq N_G(K)$), then KN is a subgroup.*
- *Size formula: $|KN| = \frac{|K||N|}{|K \cap N|}$.*

Theorem 3.2 (Second Isomorphism Theorem)**.** *Let $K \leq G$ and $N \trianglelefteq G$. Then $K \cap N \trianglelefteq K$, and*

$$\frac{KN}{N} \cong \frac{K}{K \cap N}.$$

Proof. Consider the natural projection $\pi : G \to G/N$. Restrict it to K, i.e., $\phi = \pi \mathrm{res}_K : K \to G/N$. The image of ϕ is $\{kN : k \in K\} = KN/N$. The kernel of ϕ is $\{k \in K : kN = N\} = \{k \in K : k \in N\} = K \cap N$. By the First Isomorphism Theorem, $K/\ker(\phi) \cong \mathrm{Im}(\phi)$. □

3.2 Sylow Definitions

Definition 3.3 (p-group)**.** *Let p be a prime. A group G is a p-group if every element has order a power of p. For finite groups, this is equivalent to $|G| = p^k$.*

Definition 3.4 (Sylow p-subgroup)**.** *Let $|G| = p^n m$ where $p \nmid m$. A subgroup $P \leq G$ is called a **Sylow p-subgroup** if $|P| = p^n$.*

$$Syl_p(G) = \{P \leq G : P \text{ is a Sylow } p\text{-subgroup}\}.$$

4 Lecture 4: The Sylow Theorems (Jan 21)

Theorem 4.1 (**The Sylow Theorems**). *Let G be a finite group of order $p^n m$ where $p \nmid m$.*

1. ***Existence:*** *$Syl_p(G) \neq \emptyset$. (There exists a subgroup of order p^n).*
2. ***Conjugacy:*** *Any two Sylow p-subgroups are conjugate in G. That is, if $P, Q \in Syl_p(G)$, $\exists g \in G$ such that $gPg^{-1} = Q$.*
3. ***Number:*** *Let $n_p = |Syl_p(G)|$. Then:*
   - $n_p \equiv 1 \pmod{p}$.
   - $n_p \mid m$ *(equivalently $n_p \mid |G|$).*

Proof. Let p be a prime such that $p \,\Big|\, |G|$. Define the set of all p-subgroups:

$$\Sigma = \{H : H \leq G, |H| = p^n \text{ for some } n > 0\}.$$

Define Ω as the set of maximal elements in Σ under inclusion:

$$\Omega = \{H : H \in \Sigma, \text{ there is no } K \in \Sigma \text{ such that } H \subsetneq K\}.$$

We let G act on the set of subgroups $\{H : H \leq G\}$ by conjugation. Since conjugation is an isomorphism ($H^g \simeq H$), it preserves the order of subgroups. Thus:

$$H \in \Sigma \iff H^g \in \Sigma \quad \text{and} \quad H \in \Omega \iff H^g \in \Omega.$$

Therefore, G acts on Ω by $g \cdot H = H^g$.

Claim 1: Let $H \in \Omega$. Consider the action of H on Ω by conjugation. Then H is the *unique* fixed point of this action.

Proof of Claim:

- **Existence:** It is trivial to see that H is a fixed point, as $H^h = hHh^{-1} = H$ for any $h \in H$.
- **Uniqueness:** Let $K \in \Omega$ be a fixed point of the action of H.

  $$K \text{ is fixed by } H \iff K^h = K \quad \forall h \in H \iff H \leq N_G(K).$$

  Since $K \trianglelefteq N_G(K)$, and $H \leq N_G(K)$, the product HK is a subgroup of $N_G(K)$, and thus $HK \leq G$. Using our counting lemma:

  $$|HK| = \frac{|H||K|}{|H \cap K|}.$$

  Since $H, K \in \Omega$, their orders are powers of p. Thus $|HK|$ is a power of p, meaning $HK \in \Sigma$.

  We have $H \leq HK$. Since H is maximal in Σ (by definition of Ω) and $HK \in \Sigma$, it must be that $H = HK$. Similarly, $K \leq HK$ implies $K = HK$. Therefore, $H = K$.

Claim 2: Size of Ω Modulo p

Statement: $|\Omega| \equiv 1 \pmod{p}$.

Proof of Claim: Let $H \in \Omega$. We decompose Ω into disjoint orbits under the action of H.

$$\Omega = \{H\} \cup \bigcup_{K \neq H} \mathrm{Orb}_H(K).$$

By Claim 1, H is the only fixed point (an orbit of size 1). For any $K \in \Omega$ with $K \neq H$, the stabilizer of K in H is a proper subgroup, so by the Orbit-Stabilizer theorem for p-groups, $|\mathrm{Orb}_H(K)|$ is divisible by p.

$$|\Omega| = 1 + \sum(\text{multiples of } p) \implies |\Omega| \equiv 1 \pmod{p}.$$

Claim 3: Transitivity (Conjugacy of Elements in Ω)

Statement: Any two elements of Ω are conjugate.
Proof of Claim: Suppose for contradiction that $H, K \in \Omega$ are not conjugates. Let $\theta_1 = \mathrm{Orb}_G(H)$ and $\theta_2 = \mathrm{Orb}_G(K)$. Since they are distinct orbits, $\theta_1 \cap \theta_2 = \emptyset$.

Consider the action of H on these sets:

- On θ_1: $H \in \theta_1$. By Claim 1, H is the unique fixed point of H in Ω. Since $\theta_1 \subseteq \Omega$, H is the unique fixed point in θ_1. Thus, $|\theta_1| \equiv 1 \pmod{p}$.
- On θ_2: Since $H \in \theta_1$ and $\theta_1 \cap \theta_2 = \emptyset$, we have $H \notin \theta_2$. Therefore, θ_2 contains *no* fixed points under the action of H (because the only fixed point in all of Ω is H). Thus, every orbit of H inside θ_2 has size divisible by p. This implies $|\theta_2| \equiv 0 \pmod{p}$.

By symmetry, we can swap the roles of H and K. Running the same argument with K acting on the sets implies $|\theta_2| \equiv 1 \pmod{p}$ and $|\theta_1| \equiv 0 \pmod{p}$.

This results in a contradiction (e.g., $|\theta_1| \equiv 1$ and $|\theta_1| \equiv 0$). Thus, $\theta_1 = \theta_2$, so H and K are conjugate.

Corollary: Since there is only one orbit, $\Omega = \theta_1$. Thus $|\Omega| = |\theta_1| \equiv 1 \pmod{p}$.

Claim 4: Divisibility

Statement: $|\Omega| \,\Big|\, |G|$.
Proof of Claim: Let $H \in \Omega$. Since Ω is exactly the orbit of H under conjugation (from Claim 3), the size of the orbit is the index of the stabilizer:

$$|\Omega| = [G : \mathrm{Stab}_G(H)] = [G : N_G(H)] = \frac{|G|}{|N_G(H)|}.$$

Thus $|\Omega|$ divides $|G|$.

Claim 5: Identification with Sylow Subgroups

Statement: $\Omega = Syl_p(G)$.
Proof of Claim: We know $Syl_p(G) \subseteq \Omega$ because Sylow subgroups are maximal p-subgroups by definition. We must show the reverse: every $H \in \Omega$ is a Sylow p-subgroup. Let $H \in \Omega$. Assume for contradiction that $H \notin Syl_p(G)$. Let $|G| = p^s m$ where $p \nmid m$. Since H is a p-group but not Sylow, $|H| = p^t$ with $t < s$.
Consider the normalizer $N_G(H)$. From Claim 2, we know $|\Omega| = [G : N_G(H)] \equiv 1 \pmod{p}$. Therefore, p does *not* divide $[G : N_G(H)]$. Since $|G| = [G : N_G(H)] \cdot |N_G(H)|$, all factors of p in $|G|$ must reside in $|N_G(H)|$. Thus, $|N_G(H)|$ is divisible by p^s.
Consequently, the index $[N_G(H) : H] = \frac{|N_G(H)|}{|H|} = \frac{(\text{multiple of } p^s)}{p^t}$ is divisible by p (since $s > t$).
By Cauchy's Theorem applied to the quotient group $N_G(H)/H$, there exists a subgroup of order p, say $\bar{K} \leq N_G(H)/H$. By the Correspondence Theorem, there exists a subgroup K such that $H \leq K \leq N_G(H)$ with $|K| = p \cdot |H| = p^{t+1}$.
Since K is a p-group, $K \in \Sigma$. However, $H \subsetneq K$, which contradicts the definition of H as a maximal element in Ω.
Therefore, H must already be of order p^s. Thus $\Omega = Syl_p(G)$. $\square$

5 Lecture 5: Normal Series & Solvability (Jan 23)

5.1 Normal and Subnormal Series

Definition 5.1. *A **subnormal series** of a group G is a chain of subgroups:*

$$\{1\} = H_0 \leq H_1 \leq \cdots \leq H_k = G$$

*such that $H_i \trianglelefteq H_{i+1}$ for all i. The quotient groups H_{i+1}/H_i are called the **factors** of the series.*

Definition 5.2 (Composition Series)**.** *A subnormal series is a **composition series** if all factors H_{i+1}/H_i are **simple** groups (non-trivial groups with no normal subgroups other than $\{1\}$ and themselves).*

Theorem 5.3 (Jordan-Hölder)**.** *Any two composition series of a finite group G are equivalent. That is, they have the same length, and their factors are isomorphic (up to reordering).*

5.2 Solvable Groups

Definition 5.4. *A group G is **solvable** if it has a subnormal series where all factors H_{i+1}/H_i are **abelian**.*

Definition 5.5 (Commutator)**.** *The commutator of g, h is $[g, h] = ghg^{-1}h^{-1}$. The **Derived Subgroup** (or Commutator Subgroup) G' or $[G, G]$ is the subgroup generated by all commutators.*

Proposition 5.6. *G/N is abelian if and only if $[G, G] \leq N$. Thus, $[G, G]$ is the smallest normal subgroup such that the quotient is abelian.*

Remark 5.7 (Derived Series)**.** *Define $G^{(0)} = G$, $G^{(1)} = [G, G]$, and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. G is solvable if and only if the derived series terminates at $\{1\}$ (i.e., $G^{(n)} = \{1\}$ for some n).*

6 Lecture 6: Solvable and Derived Series (Jan 26)

Lemma 6.1. *Let $\alpha \in Aut(G)$ then $\alpha([g,h]) = [\alpha(g), \alpha(h)]$.*

Lemma 6.2. *For $N \trianglelefteq G$ then G/N is abelian iff $[G,G] \leq N$.*

Theorem 6.3. *If $N \trianglelefteq G$ and $g, h \in G$, then $[gN, hN] = [g,h]N$. Additionally, gN and hN commute in G/N iff $[g,h]N = N$ iff $[g,h] \in N$.*

Definition 6.4. *G is solvable iff there is a subnormal series $\langle H_i : 0 \leq i \leq t\rangle$ (i.e. $\{1\} = H_0 \leq H_1 \leq \cdots \leq H_t = G$ with $H_i \trianglelefteq H_{i+1}$) such that $[H_{i+1}, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$. Equivalently, each factor H_{i+1}/H_i is abelian.*

Theorem 6.5. *The following are true*
1. *If G solvable and $K \leq G$ then K is solvable*
2. *If G is solvable and $N \trianglelefteq G$ then G/N is solvable.*
3. *For $N \trianglelefteq G$, G solvable iff both N and G/N are solvable*

Proof. We prove the previous theorems

1. Let $\langle H_i : 0 \leq i \leq t\rangle$ be a series in G such that $[H_{i+1}, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$. We can let $H_i' = K \cap H_i$ for $0 \leq i \leq t$ and we need to verify $\langle H_i' : 0 \leq i \leq t\rangle$ is a solvable series in K.

2. Let $\langle H_i : 0 \leq i \leq t\rangle$ be a series in G such that $[H_{i+1}, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$. Let $H_i' = \phi_N[H_i] = H_iN/N$. Verify it is solvable.

3. The forward direction follows from (1) and (2). For the reverse direction, assume N solvable and G/N solvable. Then there is a subnormal series $\langle N_i : 0 \leq i \leq s\rangle$ in N with each N_{i+1}/N_i abelian, and a subnormal series $\langle \overline{H_j} : 0 \leq j \leq t\rangle$ in G/N with each $\overline{H_{j+1}}/\overline{H_j}$ abelian. By the correspondence theorem, each $\overline{H_j}$ corresponds to a subgroup H_j of G with $N \leq H_j$, and $H_j \trianglelefteq H_{j+1}$ with $H_{j+1}/H_j \cong \overline{H_{j+1}}/\overline{H_j}$ abelian. Concatenating gives

$$\{1\} = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_s = N = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_t = G$$

as a subnormal series with abelian factors.

□

Definition 6.6. *Given G, the **derived series** of G is given by $G^{(0)} = G$ and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$ for $i \geq 0$.*

Theorem 6.7. *G is solvable iff there exists $n \geq 0$ such that $G^{(n)} = \{1\}$.*

Proof. For the backward direction, note that by construction $G^{(i)}/G^{(i+1)}$ is abelian. Setting $H_j = G^{(n-j)}$ yields a subnormal series $\{1\} = H_0 \leq H_1 \leq \cdots \leq H_n = G$ with abelian factors, so G is solvable.
For the forward direction, let $\langle H_j : 0 \leq j \leq n \rangle$ be a subnormal series in G with H_{j+1}/H_j abelian. We show by induction on i that $G^{(i)} \leq H_{n-i}$: for $i = 0$, $G^{(0)} = G = H_n$. Assuming $G^{(i)} \leq H_{n-i}$, since H_{n-i}/H_{n-i-1} is abelian, $[H_{n-i}, H_{n-i}] \leq H_{n-i-1}$, so $G^{(i+1)} = [G^{(i)}, G^{(i)}] \leq [H_{n-i}, H_{n-i}] \leq H_{n-i-1}$. Hence $G^{(n)} \leq H_0 = \{1\}$. $\square$

Theorem 6.8. *Suppose G abelian and simple, let $a \in G$ and $a \neq 1$ then $G = \langle a \rangle$ and $|a|$ is prime.*

Theorem 6.9. *Let G be simple and let $\langle H_i : 0 \leq i \leq t \rangle$ be subnormal series in G. Then there is $n < t$ such that $H_i = 1$ for $0 \leq i \leq n$ and $H_i = G$ for $n < i \leq t$.*

7 Lecture 7: Nilpotent Groups and Free Groups (Jan 28)

7.1 Nilpotent Groups

Definition 7.1. *For $H, K \leq G$, $[H, K] = \langle [h, k] : h \in H, k \in K \rangle$*

Remark 7.2. *$G^{(i)}$ is characteristic in G for all i; in particular $G^{(i)} \trianglelefteq G$.*

Lemma 7.3. *If $N \trianglelefteq G, N \leq H \leq G$ then*

$$\frac{H}{N} \leq Z\left(\frac{G}{N}\right) \iff [G, H] \leq N.$$

Definition 7.4. *A normal series in G, $\langle H_i : 0 \leq i \leq t \rangle$ is a central series if $\frac{H_{i+1}}{H_i} \leq Z\left(\frac{G}{H_i}\right)$ for all $0 \leq i \leq t-1$. Equivalently, $[G, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$.*

Definition 7.5. *G is nilpotent iff G has a central series.*

Remark 7.6. *Nilpotent groups are solvable.*

Definition 7.7. *Let G be a group*

1. *The descending central series is the sequence of subgroup given by $\gamma_1(G) = G$ and $\gamma_{i+1}(G) = [G, \gamma_i(G)]$ for $i \geq 1$.*
2. *The ascending central series is the sequence of subgroup given by $Z_0(G) = 1$ and*

$$\frac{Z_{n+1}(G)}{Z_n(G)} = Z\left(\frac{G}{Z_n(G)}\right).$$

Remark 7.8. *By induction, $Z_n(G)$ and $\gamma_n(G)$ are characteristic in G. and $Z_n(G) \trianglelefteq G$ and $\gamma_n(G) \trianglelefteq G$ for all $n \geq 0$.*

Remark 7.9. *For any G and $N \trianglelefteq G, [G, N] \leq N$ because for any $g \in G, n \in N$ we have $[g, n] = gng^{-1}n^{-1}$ and $n^g \in N$ since $N \trianglelefteq G$. So $[G, N] \leq N$.*

Theorem 7.10. *G is nilpotent iff there is a $n \geq 0$ such that $Z_n(G) = G$ iff there is $n \geq 0$ such that $\gamma_{n+1}(G) = \{1\}$. Moreover, the least n such that $Z_n(G) = G$ is the least n such that $\gamma_{n+1}(G) = \{1\}$.*

Proof. Take any central series then the decreasing and ascending central series will grow at least as fast as the arbitrary central series. □

7.2 Free Groups

Example 7.11. *Consider* $\mathbb{Z} = (\mathbb{Z}, +)$. *The element* 1 *is a generator:* $\mathbb{Z} = \langle 1 \rangle$. *The* ***universal property of*** $\mathbb{Z}$ *states: for any group* G *and any* $g \in G$, *there is a unique homomorphism* $\phi : \mathbb{Z} \to G$ *such that* $\phi(1) = g$.

Proof. Define $\phi(n) = g^n$; this is a homomorphism. Uniqueness follows because any homomorphism ϕ with $\phi(1) = g$ must satisfy $\phi(n) = \phi(1)^n = g^n$. □

Intuitively, 1 *is "free" because there are no relations it must satisfy: whatever* g *we map* 1 *to, the assignment extends consistently to all of* $\mathbb{Z}$.

Definition 7.12. F *is a* ***free group on*** 2 ***elements*** *if*

1. *There exist* $a, b \in F$ *with* $F = \langle a, b \rangle$.
2. *For every group* G *and all* $g, h \in G$, *there is a unique homomorphism* $\phi : F \to G$ *such that* $\phi(a) = g$ *and* $\phi(b) = h$.

8 Lecture 8: Free Groups

Theorem 8.1. *For any set X, there are a group F and an injective function $i : X \to F$ such that*

1. *For any group G and any function $f : X \to G$ there is a unique homomorphism $\phi : F \to G$ such that $\phi(i(x)) = f(x)$ for all $x \in X$.*

$$\begin{tikzcd} X \arrow[r, hook, "i"] \arrow[dr, "f"'] & F \arrow[d, dashed, "\phi"] \\ & G \end{tikzcd}$$

   The diagram commutes: $\phi \circ i = f$. The homomorphism ϕ is unique.

2. *F is generated by $i[X]$*

Proof. A **word** (on alphabet X) is a (possibly empty) finite sequence of pairs from $X \times \mathbb{Z}$. We write $x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ for the word $\langle (x_1, n_1), (x_2, n_2), \ldots, (x_k, n_k) \rangle$.

Example 8.2. *Let $X = \{x, y, z\}$. Then $w = x^0 x^{15} y^{-23} z^{27}$ is a word.*

Let W denote the set of all words. Given a word, we can possibly perform a **reduction step**:

1. If the word contains an entry $(x, 0)$, remove it.
2. If the word contains successive entries (x, n) and (x, m) (same x), replace them with $(x, n + m)$.

Example 8.3. $y^{15} x^3 x^{-2} x^{-1} y^2 z^5 \to y^{15} x^1 x^{-1} y^2 z^5 \to y^{15} x^0 y^2 z^5 \to y^{15} y^2 z^5 \to y^{17} z^5$.

Let R be the set of all **reduced** words (those on which no further reductions can be performed).
For any word w, there is a unique reduced word w' such that w can be transformed into w' by a finite sequence of reductions.
We will not be able to show this confluence property purely combinatorially; instead, we produce a group F such that every word is equivalent to a unique reduced word.

Given $x \in X$, define $f_x : R \to R$ by cases:

1. If w does not begin with x^m for any m, then $f_x(w) = xw$.
2. If $w = x^{-1} v$, then $f_x(w) = v$.
3. If $w = x^n v$ for $n \neq -1$, then $f_x(w) = x^{n+1} v$.

Similarly define $g_x : R \to R$ by

1. If w does not begin with x^n for any n, then $g_x(w) = x^{-1} w$.
2. If $w = x^1 v$, then $g_x(w) = v$.
3. If $w = x^n v$ for $n \neq 1$, then $g_x(w) = x^{n-1} v$.

One checks that $f_x \circ g_x = g_x \circ f_x = 1_R$.
For each $x \in X$, $f_x \in \mathrm{Sym}_R$ (the group of permutations of R), and $g_x = f_x^{-1}$. Let F be the subgroup of Sym_R generated by $\{f_x : x \in X\}$; its elements are products $f_{x_1}^{n_1} \cdots f_{x_t}^{n_t}$ with $x_i \in X$, $n_i \in \mathbb{Z}$.
Let $i : X \to F$ with $i(x) = f_x$. For any word $w = x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t}$, let $f_w = f_{x_1}^{n_1} f_{x_2}^{n_2} \cdots f_{x_t}^{n_t} \in F$.
Key Facts:

1. If w is obtained from $\overline{w}$ by a single reduction step, then $f_w = f_{\overline{w}}$.
2. If w is obtained from $\overline{w}$ by any sequence of reduction steps, then $f_w = f_{\overline{w}}$.
3. If $w \in R$ then $f_w(\varepsilon) = w$, where ε denotes the empty word.
4. For any word w, $f_w(\varepsilon)$ is the unique reduced word to which w can be transformed by reduction steps.
5. If w_1, w_2 are reduced words and v is the unique reduction of the concatenation $w_1 w_2$, then $f_v = f_{w_1} f_{w_2}$.

Let F' be the group whose underlying set is R, with operation "concatenate and reduce". Then $F \simeq F'$ via $f_v \mapsto v$. To finish, let G be a group and $h : X \to G$ a function. We want to show there is a unique homomorphism $\phi : F' \to G$ with $\phi \circ i = h$.

1. If ϕ exists, then

$$\begin{aligned}\phi(x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t}) &= \phi(i(x_1))^{n_1} \phi(i(x_2))^{n_2} \cdots \phi(i(x_t))^{n_t} \\ &= h(x_1)^{n_1} h(x_2)^{n_2} \cdots h(x_t)^{n_t},\end{aligned}$$

   which determines ϕ uniquely.

2. One verifies this formula defines a homomorphism.

$\square$

Example 8.4. *Let $X = \{x, y\}$ and let G be any group with 2 generators, say $G = \langle a, b \rangle$. Let F be a free group on X, and let $\phi : F \to G$ be the unique homomorphism such that $\phi(x) = a$ and $\phi(y) = b$. Then ϕ is surjective (since G is generated by a, b), so by the First Isomorphism Theorem, $G \simeq F/\ker(\phi)$.*

9 Lecture 9: Generators and Relations (Feb 2)

Definition 9.1. *Let X be a set, and let F be a free group on X. Let $A \subseteq F$, and let N be the least normal subgroup of F containing A (equivalently, N is the normal closure of A in F).*

Theorem 9.2. *Let $\overline{F} = F/N$ and $j : X \to \overline{F}$ by $j(x) = xN$.*

$$\begin{tikzcd} F \arrow[r, "\phi_N"] & \overline{F} \\ X \arrow[u, "i"] \arrow[ur, "j"'] & \end{tikzcd}$$

The following are true:

1. *$\overline{F}$ is generated by $\{j(x) : x \in X\}$. Let $\overline{x} = j(x)$*
2. *If $a = x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t} \in A$ then $a \in N$ and $\overline{x_1}^{n_1} \overline{x_2}^{n_2} \cdots \overline{x_t}^{n_t} = \phi_N(a) = 1_{\overline{F}}$*

"Universal Property of $\overline{F}$": Let G be a group and let $h : X \to G$ such that $a = x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t} \in A$ and $h(x_1)^{n_1} h(x_2)^{n_2} \cdots h(x_t)^{n_t} = 1_G$. There is an unique homomorphism $\mathcal{T} : \overline{F} \to G$ such that

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[dr, "h"'] & \overline{F} \arrow[d, dashed, "\mathcal{T}"] \\ & G \end{tikzcd}$$

So $h = \mathcal{T} \circ j$

Proof. Since $\{\overline{x} : x \in X\}$ generates $\overline{F}$ there is at most one τ. By universal property of F, there is unique homomorphism $\phi : F \to G$ such that

$$\phi(x^1) = h(x). \qquad \text{(For all } x \in X\text{)}$$

Use property of F

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr, "h"'] & F \arrow[d, dashed, "\phi"] \\ & G \end{tikzcd}$$

Claim: By hypothesis on h, $A \subseteq \ker(\phi)$.
Let $a = x_1^{t_1} x_2^{t_2} \cdots x_k^{t_k} \in A$ then $\phi(a) = \phi(x_1)^{t_1} \phi(x_2)^{t_2} \cdots \phi(x_k)^{t_k} = h(x_1)^{t_1} h(x_2)^{t_2} \cdots h(x_k)^{t_k} = 1_G$ since $h(x_1)^{t_1} h(x_2)^{t_2} \cdots h(x_k)^{t_k} = 1_G$. Thus, $a \in \ker(\phi)$.

By choice of N, $N \leq \ker(\phi)$, we'll attempt to define $\tau : \overline{F} \to G$ by $\tau(wN) = \phi(w)$ for $w \in F$.
τ is well defined because $w_1 N = w_2 N \implies w_1^{-1} w_2 \in N \implies w_1^{-1} w_2 \in \ker(\phi) \implies \phi(w_1)^{-1} \phi(w_2) = 1 \implies \phi(w_1) = \phi(w_2)$. Showing τ is a homomorphism is straightforward.

$$\begin{tikzcd}
X \arrow[r, "i"] \arrow[dr, "h"'] & F \arrow[d, "\phi_N"] \arrow[dr, "\phi"] & \\
& \overline{F} \arrow[r, dashed, "\mathcal{T}"] & G
\end{tikzcd}$$

$\square$

10 Lecture 10: Presentation of Groups (Feb 4)

Example 10.1. *Dihedral Group: D_n = symmetry group of a regular n-gon.*
Let $D_n = \langle a, b\rangle$ where a = rotation by $\frac{2\pi}{n}$ and b = reflection about the vertical axis. Then D_n is generated by a and b with the relations $a^n = 1$ and $b^2 = 1$.
We have $bab^{-1}a = 1$
Let $X = \{x, y\}$ and F be free group and $A = \{x^n, y^2, yxy^{-1}x\}$ and N = least normal subgroup $\supseteq A$. Let $\overline{x} = xN$ and $\overline{y} = yN$. Then $F/N = \langle \overline{x}, \overline{y}\rangle$ and

$$\overline{yxy^{-1}x} = \overline{yxy}^{-1}\overline{x}^{-1} = 1_{\overline{F}}.$$

There is a unique homomorphism $\tau : F/N \to D_n$ such that $\tau(\overline{x}) = a$ and $\tau(\overline{y}) = b$ as $D_n = \langle a, b\rangle$ so τ is surjective. So $|F/N| \geq 2n$.
Claim: $|F/N| \leq 2n$ (so τ is isomorphism)
Proof: Using relations from before, we can simply write any expression in $\overline{x}, \overline{y}$ down to $\overline{x}^s\overline{y}^t$ for $0 \leq s \leq n$ and $0 \leq t \leq 2$.

11 Lecture 11: Category Theory (Feb 6)

11.1 Categories

Example 11.1. *1. Category of sets, if we have two sets A and B then $A \to B$ is a function $f : A \to B$*
2. Category of groups, if we have two groups G and H then $G \to H$ is a homomorphism $\phi : G \to H$
3. Category of topological spaces, if we have two topological spaces X and Y then $X \to Y$ is a continuous function $f : X \to Y$

Definition 11.2. *A category $\mathcal{C}$ consists of the following data:*
1. *A collection of objects $\mathrm{Ob}(\mathcal{C})$.*
2. *A collection of morphisms $\mathrm{Mor}(\mathcal{C})$.*
3. *For each morphism f, objects $\mathrm{dom}(f)$ and $\mathrm{cod}(f)$ with $f : \mathrm{dom}(f) \to \mathrm{cod}(f)$.*
4. *A partial composition operation: for composable $f, g \in \mathrm{Mor}(\mathcal{C})$ (i.e. $\mathrm{cod}(f) = \mathrm{dom}(g)$), a morphism $g \circ f \in \mathrm{Mor}(\mathcal{C})$.*
5. *An identity morphism $1_A : A \to A$ for each object $A \in \mathrm{Ob}(\mathcal{C})$.*

These satisfy the usual unit and associativity laws (see lemma below).

Lemma 11.3. *The following are true:*
1. *If $a \xrightarrow{f} b$ then $f1_a = f$ and $1_b f = f$.*
2. *If $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$ then $h(gf) = (hg)f$*

Example 11.4. *Posets: $\mathbb{P}$ is a poset with $\leq$ binary relation such that*
1. *$a \leq a$ for all $a \in \mathbb{P}$*
2. *$a \leq b$ and $b \leq a \implies a = b$*
3. *$a \leq b$ and $b \leq c \implies a \leq c$*

Given poset $\mathbb{P}$, make "poset category" with objects the elements of $\mathbb{P}$. There is only one morphism $a \to b$ if $a \leq b$ and no morphism if $a \not\leq b$.

Definition 11.5. *Let $\mathcal{C}$ be a category. An object a is **initial** if for every object b there is exactly one arrow $a \to b$.*

Definition 11.6. *An arrow $a \xrightarrow{f} b$ in $\mathcal{C}$ is an **isomorphism** iff there exists $b \xrightarrow{g} a$ such that $gf = 1_a$ and $fg = 1_b$.*

Lemma 11.7. *If a, b are initial objects in $\mathcal{C}$, they are uniquely isomorphic.*

Proof. Let $a \xrightarrow{f} b$ and $b \xrightarrow{g} a$ be the unique arrows. Then $gf : a \to a$ and $fg : b \to b$. Since a is initial, there is only one arrow $a \to a$, namely 1_a, so $gf = 1_a$. Similarly $fg = 1_b$. □

Definition 11.8. *An object b is **terminal** iff for every a there is a unique arrow $a \to b$, equivalently, b is initial in $\mathcal{C}^{op}$.*

Lemma 11.9. *Any two terminal objects are uniquely isomorphic.*

Definition 11.10. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A* ***functor*** $F : \mathcal{C} \to \mathcal{D}$ *assigns to each object a of $\mathcal{C}$ an object $F(a)$ of $\mathcal{D}$, and to each arrow $a \xrightarrow{f} b$ an arrow $F(a) \xrightarrow{F(f)} F(b)$, such that $F(1_a) = 1_{F(a)}$ and $F(gf) = F(g)F(f)$ whenever $g \circ f$ is defined.*

12 Lecture 12: Ring Theory (Feb 8)

Definition 12.1. *A ring is a set R with two binary operations $+$ and $\times$ such that*
1. *$(R, +)$ is an abelian group*
2. *$(R, \times)$ is associative*
3. *$a \times (b + c) = a \times b + a \times c$ and $(b + c) \times a = b \times a + c \times a$ for all $a, b, c \in R$*

Example 12.2. *The following are rings*
1. $\mathbb{Z}, 2\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
2. $M_2(\mathbb{Z})$
3. $\{f : \mathbb{R} \to \mathbb{R}$ *continuous*$\}$
4. $\{f : \mathbb{R} \to \mathbb{R}$ *differentiable*$\}$

Definition 12.3. *R is **commutative** iff $rs = sr$ for all $r, s \in R$.*

Definition 12.4. *R is unital iff there exists $1_R \in R$ such that $1_R r = r 1_R = r$ for all $r \in R$*

Definition 12.5. *Let R, S be rings. A ring homomorphism from R to S if $\phi : R \to S$ satisfy:*
1. *$\phi(r + s) = \phi(r) + \phi(s)$ for all $r, s \in R$*
2. *$\phi(rs) = \phi(r)\phi(s)$ for all $r, s \in R$*
3. *$\phi(1_R) = 1_S$*

Definition 12.6. *Let S be a subring of R if S is a ring and the inclusion map $i : S \to R$ is a ring homomorphism.*

Definition 12.7. *Let $\phi : R \to S$ be a ring homomorphism. Then*
1. *$\ker(\phi) = \{r \in R : \phi(r) = 0\}$ is an* ideal *of R.*
2. *$\phi(R) = \{s \in S : \exists r \in R, \phi(r) = s\}$ is a subring of S.*

Remark 12.8. *In general, $\ker(\phi)$ is not a (unital) subring of R, since it does not contain 1_R unless $\phi = 0$. However, it is always an ideal.*

Definition 12.9. *Let R be a ring. An **ideal** of R is a subset $I \subseteq R$ such that*
1. *$I \leq (R, +)$ (i.e. I is an additive subgroup).*
2. *For all $a \in R$ and $b \in I$, both $ab \in I$ and $ba \in I$.*

Definition 12.10. *Let I be an ideal of R then we define $r + I = \{r + i : i \in I\}$*

13 Lecture 13: Ideals and Quotient Rings (Feb 10)

Theorem 13.1 (First Isomorphism Theorem for Rings)**.** *Let R, S be rings and let $\phi : R \to S$ be a ring homomorphism. Then $\ker(\phi)$ is an ideal of R, $\mathrm{im}(\phi)$ is a subring of S, and there is a ring isomorphism*

$$\tau : \frac{R}{\ker(\phi)} \xrightarrow{\cong} \mathrm{im}(\phi), \qquad \tau : r + \ker(\phi) \mapsto \phi(r).$$

Theorem 13.2. *The following are true for any ideals I, J of R:*

1. *$I + J =$ smallest ideal containing I and J*
2. *$I \cap J =$ largest ideal contained in I and J*
3. *$IJ = \{\sum_{i=1}^{n} r_i s_i | n \in \mathbb{N}, r_i \in I, s_i \in J\} \subseteq I \cap J$*
4. *Ideals of R/I are in bijection with ideals of R that contain I*

Definition 13.3. *R is a **zero ring** if $R = \{0_R\}$.*
Note: If $1_R = 0_R$, then for any $r \in R$, $r = r \cdot 1_R = r \cdot 0_R = 0_R$, so R is a zero ring.

Definition 13.4. *R is an integral domain iff*

1. *$1_R \neq 0_R$ (R is not a zero ring)*
2. *For all $a, b \in R$*
$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

Definition 13.5. *R is a field iff*

1. *$1_R \neq 0_R$*
2. *Every element of $R \setminus \{0_R\}$ has a multiplicative inverse*

Exercise 13.6. *If R is a field then R is an integral domain.*

Definition 13.7. *Let R be a ring. Then $u \in R$ is a unit iff u has a multiplicative inverse.*
We let
$$U(R) = \{u \in R : u \text{ is a unit}\}.$$
$U(R)$ is a group under multiplication.

Definition 13.8. *An ideal I of R is principal iff $I = aR$*

Lemma 13.9. *Let R be any ring then R is the largest ideal and $\{0_R\}$ is the smallest ideal.*

Remark 13.10. *We have $R/R = \{0_R\}$ and $R/\{0_R\} \simeq R$*

Theorem 13.11. *Let R be a field and let I be an ideal then $I = 0$ or $I = R$*

Proof. Assume $I \neq 0$, let $a \in I$ and $a \neq 0$. There exist $b \in R$ such that $ab = 1 \in I$. In

any ring $1 \in I$ iff $I = R$.

$\square$

Theorem 13.12. *If $R \neq 0$ and (0) and R are the only two ideals of R, then R is a field.*

Proof. Let $a \in R$ with $a \neq 0$. Then $(a) \neq (0)$ since $a \in (a)$, so $(a) = R$. Hence $1 \in (a)$, meaning $1 = ab$ for some $b \in R$, so a has a multiplicative inverse. $\square$

Definition 13.13. *An ideal I of R is maximal iff*

1. *$I \neq R$*
2. *For all ideals $J \supseteq I$, $J = I$ or $J = R$*

I is maximal in poset $\{J : J \text{ ideal}, J \neq R\}$ ordered by $\subseteq$

Lemma 13.14. *If I is an ideal of R,*

$$I \text{ maximal} \iff R/I \text{ is a field.}$$

Proof. $R \neq I \iff 1_{R/I} \neq 0_{R/I}$. By I maximal iff there are only two ideals of R/I which are 0 and R/I. $\square$

Definition 13.15. *Ideal I is prime iff*

1. *$I \neq R$*
2. *For all $a, b \in R$, $ab \in I \implies a \in I$ or $b \in I$*

Lemma 13.16. *R/I is an integral domain iff I is a prime ideal.*

Definition 13.17. *$Spec(R) = \{I : I \text{ is a prime ideal of } R\}$*

Theorem 13.18. *Let R be a ring. I is an ideal of R, $I \neq R$. Then there is a maximal ideal J such that $I \subseteq J$.*

14 Lecture 14: Zorn's Lemma and Modules (Feb 13)

Lemma 14.1. *Zorn's Lemma: If $\mathbb{P}$ is a poset such that every chain in $\mathbb{P}$ has an upper bound, then for every element p of $\mathbb{P}$ there is a $q \geq p$ with q maximal.*

Example 14.2. *Claim: $\mathbb{Q}$ satisfies the hypothesis of Zorn's Lemma.*

Proof. Let $(I_a : a \in A)$ be a chain in $\mathbb{Q}$. That is, I_a is ideal with $I_a \neq R$ and for $a, b \in A, I_a \subseteq I_b$ or $I_b \subseteq I_a$. We need upper bound, let

$$J = \bigcup_{a \in A} I_a = \{r : \text{there is } a \in A, r \in I_a\}.$$

Verify J is an ideal and for $r, s \in J$ then $r \in I_a$ and $s \in I_b$. WLOG let $I_a \subseteq I_b$ so $r, s \in I_b$ so $r + s \in I_b \subseteq J$.
Verify $J \neq R$, easy as $1 \notin I_a$ all a so $1 \notin J$. □

14.1 Modules

Definition 14.3. *Let R be a ring. A (left) R-**module** is an abelian group $(M, +)$ equipped with a map $R \times M \to M$, $(r, m) \mapsto rm$, such that*
1. *$0_R m = 0_M$ and $1_R m = m$ for all $m \in M$.*
2. *$(r_1 + r_2)m = r_1 m + r_2 m$ and $r(m_1 + m_2) = rm_1 + rm_2$ for $r, r_i \in R$, $m, m_i \in M$.*
3. *$r(sm) = (rs)m$ for $r, s \in R$ and $m \in M$.*

Definition 14.4. *If M is a R-module, we can form "linear combinations"*

$$\sum_{i=1}^{n} r_i m_i \text{ for } r_i \in R \text{ and } m_i \in M.$$

Example 14.5. *Let $R = \mathbb{Z}$ and $(G, +)$ be an abelian group. Then G is a $\mathbb{Z}$-module.*

Definition 14.6. *If M is a R-module, a submodule of M is N such that $N \leq (M, +)$, $rn \in N$ for $r \in R$ and $n \in N$.*

Remark 14.7. *If R is a ring, we can view R as a R-module. The ideals of R are the submodules of R (viewed as a R-module).*

Definition 14.8. *Let M, N be R-modules. A R-module homomorphism is a function $\phi : M \to N$ such that*
1. *ϕ is a group homomorphism for $(M, +)$ to $(N, +)$*
2. *$\phi(rm) = r\phi(m)$ for $r \in R$ and $m \in M$*

Note: This is like linear transformation in linear algebra.

15 Lecture 15: Modules (Feb 16)

Definition 15.1. *Let M and N be R-modules then we say $\phi : M \to N$ is R-linear if*

$$\phi\left(\sum_{i=1}^{n} r_i m_i\right) = \sum_{i=1}^{n} r_i \phi(m_i) \text{ for } r_i \in R \text{ and } m_i \in M.$$

Definition 15.2. *Let $M \leq N$ (M is submodule of N), $N/M = \{n + M : n \in N\}$. We define*

1. $(n_1 + M) + (n_2 + M) = (n_1 + n_2) + M$
2. $r(n + M) = (rn) + M$

We can define the R-linear quotient map $\phi_M : N \to N/M$ by $\phi : n \mapsto n + M$ with $\ker(\phi_M) = M$.

Theorem 15.3. *First Isomorphism Theorem for Modules: Let $\phi : M_1 \to M_2$ linear, then*

$$\mathrm{im}(\phi) \simeq \frac{M_1}{\ker(\phi)}.$$

via $m + \ker(\phi) \mapsto \phi(m)$

Definition 15.4. *Given R-module M and set $X \subseteq M$, the least submodule containing X is*

$$\mathrm{Span}(X) = \left\{\sum_{i=1}^{n} r_i x_i | n \in \mathbb{N}, r_i \in R, x_i \in X\right\}.$$

We specifically define $\mathrm{Span}(\emptyset) = \{0_M\}$

Definition 15.5. *If R is a ring, $X \subseteq R$ we write $\langle X \rangle$ for the least ideal containing X*

$$\langle X \rangle = \left\{\sum_{i=1}^{n} r_i x_i | n \in \mathbb{N}, r_i \in R, x_i \in X\right\}.$$

Definition 15.6. *A ring R is a principal ideal ring iff all ideals of R are principal (a principal ideal is of form Ra for some $a \in R$).*

Definition 15.7. *R is a principal ideal domain iff R is integral domain and principal ideal ring.*

Definition 15.8. *A ring R is noetherian iff every ideal of R is finitely generated. (An ideal is finitely generated if $I = Ra_1 + \cdots + Ra_n$ for some $a_1, \ldots, a_n \in R$.)*

Theorem 15.9. *If I is an ideal of R the ideals R/I are in bijection with $\{J : J \text{ ideal of } R, J \supseteq I\}$.*
Also prime ideals of R/I correspond to prime ideals of R that contain I.

Example 15.10. *Common PID's*
1. $\mathbb{Z}$
2. $k[x]$ *for* k *a field*

Definition 15.11. *R is a Euclidean Domain iff*
1. *R is integral domain*
2. *There is a function $\phi : R \setminus \{0\} \to \mathbb{N}$ such that for all $a, b \in R$, for $b \neq 0$, there exist q, r such that $a = bq + r$ either $r = 0$ or $\phi(r) < \phi(b)$*

Example 15.12. *If $R = \mathbb{Z}$ then $\phi(n) = |n|$ is a Euclidean Domain.*

Example 15.13. *If $R = k[x]$ for k a field then $\phi(f) = \deg(f)$ is a Euclidean function.*

Theorem 15.14. *If R is a Euclidean Domain then R is a PID.*

Proof. Let R be euclidean domain and let I be the ideal of R. If $I = 0$ then we're done. If $I \neq 0$ consider $\{\phi(r) | r \in I, r \neq 0\} \subseteq \mathbb{N}$. Let n be the minimal element and let $a \in I \setminus \{0\}$ such that $\phi(a) = n$. As $a \in I, Ra \subseteq I$. Let $b \in I$, find q, r such that $b = aq + r$ either $r = 0$ or $\phi(r) < n$. We have $r = b - aq \in I$. So $r = 0$ and $b \in Ra$. $\square$

Lemma 15.15. *Facts about Integral Domains: Let R be an integral domain.*
1. *If $a \neq 0$ and $ab = ac$ then $a(b - c) = 0$*
2. *$Ra = Rb$ iff a and b are associates*

Lemma 15.16. *Let R be any ring, the following are true about units*
1. *u unit iff u has a multiplicative inverse*
2. *Units of R form a group under multiplication*
3. *$a, b \in R$ are called associates iff there is an unit u such that $au = b$.*

Example 15.17. *$\mathbb{Z}[x]$ is not a PID. Consider the ideal $\langle 2, x \rangle = \{f :$ constant term of f is even$\}$.*

Theorem 15.18. *R Noetherian iff $R[x]$ is Noetherian.*

16 Lecture 16: Integral Domains

Definition 16.1. *Let $p \in R$*

1. *p is prime if and only if p is a nonzero non-unit and for all $r, s \in R$, $p|rs$ implies $p|r$ or $p|s$.*
2. *p is irreducible if and only if p is a nonzero non-unit and for all $r, s \in R$ and $p = rs$ then one of r or s is a unit.*

Lemma 16.2. *The following are true*

1. *As R is an ID, for any $a \in R$, a nonzero non-unit iff $(a) \notin \{0, R\}$*
2. *If p is prime, then (p) is a prime ideal*
3. *p prime implies p is irreducible.*
4. *If R PID then prime = irreducible.*
5. *In any ID, (0) is a prime ideal.*

Definition 16.3. *R is a unique factorization domain (UFD) iff*

1. *R is an integral domain*
2. *Every non-zero nonunit in R is a product $u_1, ..., u_n$, $n > 0, u_i$ irreducible.*
3. *If $m, n > 0$ then $u_1...u_n = v_1...v_m$ and u_i, v_j irreducible, then $m = n$ and u_i are associates and permutations of v_j*

Recall 16.4. *An R-module M is finitely generated (fg) if there are $m_1, ..., m_n \in M$ such that $M = \mathrm{Span}(m_1, ..., m_n)$.*

Definition 16.5. *An R-module N is Noetherian iff every submodule of N is finitely generated.*
Equivalently, Ring R is Noetherian iff R is Noetherian as an R-module that is all ideals are finitely generated.

Example 16.6. *Non-Noetherian Rings:*

$$R = \bigcup_{n=1}^{\infty} \mathbb{Z}[x_1, ..., x_n].$$

Consider the ideal

$$I = \langle x_1, x_2, ... \rangle = \{f \in R : f \text{ has no constant term}\} = \{f : f(0, ...) = 0\}.$$

Proof. If I is finitely generated there is $h_1, ..., h_n \in I$ such that

$$I = Rh_1 + \cdots + Rh_n = (h_1) + \cdots + (h_n).$$

Each h_i has finite x_j's in it so find N so large that x_N does not appear in any h_i.
As $x_N \in I$ there are $f_1, ..., f_n \in R$ such that $x_N = \sum_{i=1}^{n} f_i h_i$. Substitute

$$x_j = \begin{cases} 0 & \text{if } j < N \\ 1 & \text{if } j \geq N \end{cases}.$$

Then we conclude $1 = 0$ which is a contradiction. $\square$

Theorem 16.7. *Let M be a R-module then the following are equivalent*

1. *M is Noetherian*
2. *For any increasing change of submodules $N_0 \subseteq N_1 \subseteq N_2 \subseteq \cdots$ of M, the chain is eventually constant.*
3. *For any nonempty set X of submodules of M, there is some $N \in X$ which is maximal under inclusion.*

Proof. (1) $\implies$ (2): Consider $N_\infty = \bigcup_{i=0}^{\infty} N_i = \{m : \exists i \in \mathbb{N}, m \in N_i\}$.
Verify that N_∞ is a submodule of M, as M is noetherian, N_∞ is finitely generated say $N_\infty = \operatorname{Span}(X)$ with $X \subseteq N_\infty$ finite. Find i such that $X \subseteq N_i$ then $N_\infty = \operatorname{Span}(X) \subseteq N_i \subseteq N_j \subseteq N_\infty$ for $j \geq i$.

(2) $\implies$ (1): AFSOC there is $N \leq M$ with N not finitely generated. Take $n_0 \in N$ and $n_0 \neq 0$. Choose $n_1 \in N \setminus Rn_0$ and generally

$$n_k \subseteq N \setminus \left(\sum_{i<k} Rn_i \right).$$

In the end, let $N_i = \sum_{j<i} Rn_j$ then $N_0 < N_1 < N_2 < \cdots$, a contradiction. $\square$

17 Lecture 17: PID (Feb 19)

Lemma 17.1. *If R PID then*
1. *R is Noetherian*
2. *0 is a prime ideal and 0 is maximal iff R is a field*
3. *For $a \neq 0, (a)$ prime iff a prime iff (a) maximal iff a irreducible*

Theorem 17.2. *If R is a PID then R UFD*

Claim 17.3. *Every nonzero non-unit $a \in R$ is a finite product of irreducibles.*

Proof. If not, choose a that is not a finite product of irreducibles such that (a) is maximal under inclusion, a not finite product of irreducibles so a is not irreducible.
So $a = bc$ where b, c not units. So $(a) \subsetneq (b)$ and $(a) \subsetneq (c)$ so b, c are finite products of irreducibles as (a) is maximal under inclusion. Contradiction so a is a finite product of irreducibles. □

Claim 17.4. *If $u_1...u_n = v_1...v_n$ with u_i, v_j irreducible then $m = n$ and u_i, v_j are the same up to permutations and associates.*

Proof. u_1 divides $v_1...v_m$. As R PID, u_1 is prime so $u_1|v_j$ for some j. As v_j is irreducible, u_i either unit or an associate of v_j. As u_i irreducible, u_i not a unit. So u_1 and v_j are associates. Say $u_1 = x \cdot v_j$ for x unit. Then

$$xu_2...u_n = v_1...v_{j-1}v_{j+1}...v_m.$$

Finish by induction. □

Corollary 17.5. *If k field, $k[x]$ is a UFD.*

Lemma 17.6. *If R PID and $p \in R$ irreducible then $R/(p)$ is a field.*

Theorem 17.7. *(Hilbert's Basissatz Theorem) If R Noetherian ring then $R[x]$ is Noetherian ring.*

Proof. Let I be ideal of $R[x]$. For each n let

$$J_n = \left\{ r \in R : \exists c_0, ..., c_{n-1} \in R, rx^n + \sum_{i=0}^{n-1} c_i x^i \in I \right\}.$$

J_n is an ideal of R.
Claim: $J_n \subseteq J_{n+1}$
Let $r \in J_n$ say $rx^n + \sum_{i=0}^{n-1} c_i x^i \in I$, as I ideal, $x(rx^n + ...) \in I$ so $r \in J_{n+1}$.

As R is noetherian, there is N such that $J_n = J_N$ for all $n \geq N$.
For each $n \leq N$, as R noetherian, J_n is finitely generated ideal of R. For each $n \leq N$,

choose a finite set $F_n \subseteq I$ such that if $\deg(f) = n$ then $f \in F_n$ and $\{$leading coeff of f : $f \in F_n\}$ generates J_n.
Let $F = \bigcup_{n \leq N} F_n$ and claim that F generates I (as an ideal of $R[x]$). Show by induction on $\deg(h)$ that every $h \in I$ is an $R[x]$-linear combination of elements of F.
(Base Case) $h = 0$ is trivial. More generally, suppose $\deg(h) = n \leq N$ and $h = rx^n + \sum_{i<n} c_i x^i$ where $r \in J_n$. Find $\lambda_1, ..., \lambda_t \in R$ and $g_1, ..., g_t \in F_n$ such that r is the leading coefficient of $\sum_{i=1}^{t} \lambda_i g_i$. Then $h - \sum_{i=1}^{t} \lambda_i g_i \in I$ has strictly lower degree, so we finish by induction.
(Induction Step) Let $h \in I$ with $\deg(h) = n > N$, $h = rx^n + \sum_{i<n} c_i x^i$ with $r \in J_n = J_N$. Find $g'_1, ..., g'_s \in F_N$ and $\lambda_1, ..., \lambda_s \in R$ such that r is the leading coefficient of $\sum_{j=1}^{s} \lambda_j g'_j$ (whose leading term is rx^N). Then $h - \sum_{j=1}^{s} \lambda_j x^{n-N} g'_j \in I$ has degree $< n$, so we are done by induction. $\square$

18 Lecture 18: Unique Factorization Domains (Feb 23)

Theorem 18.1. *R UFD $\implies$ $R[x]$ UFD*

Definition 18.2. *Field of Fractions: Given any integral domain R, constructing its "field of fractions" as $X = \{(a,b) \in R \times R : b \neq 0\}$.*
Define binary relation $\sim$ on X by

$$(a_1, b_1) \sim (a_2, b_2) \iff a_1 b_2 = a_2 b_1.$$

Definition 18.3. *Define $\frac{a}{b}$ to be the $\sim$-equivalence class of (a,b).*
The fractions form a ring F with $1_F = \frac{1}{1}$ and $0_F = \frac{0}{1}$. Every element has an inverse, so F is a field.

Remark 18.4. *If R is integral domain and F is a field of fractions then $F[x]$ is a PID.*

Definition 18.5. *Let R be an integral domain and let $a_1, ..., a_n \in R, a_i \neq 0$. A greatest common divisor (gcd) for $a_1, ..., a_n$ is $a \in R$ such that*
1. *$a | a_i$ for $i = 1, ..., n$*
2. *If $b | a_i$ for $i = 1, ..., n$ then $b | a$*

Remark 18.6. *If $\gcd(a_1, ..., a_n)$ exists, it is unique up to associates.*

Lemma 18.7. *If R UFD, then gcd's exist.*

Lemma 18.8. *If R PID and $(a_1, ..., a_n) = Ra_1 + \cdots + Ra_n$ then*

$$(a_1, ..., a_n) = (d) \iff d = \gcd(a_1, ..., a_n).$$

Definition 18.9. *Let R, S be rings, $\phi : R \to S$ a ring homomorphism.*
1. *If I is an ideal of R, I^e (extension of I) = ideal generated by $\phi[I]$ in S.*
2. *If J ideal of S, J^c (contraction) of J is $\phi^{-1}[J] = \{r \in R : \phi(r) \in J\}$.*

Lemma 18.10. *If we define $\psi : R/J^c \to S/J$ by $\psi : r + J^c \mapsto \phi(r) + J$, ψ is well defined injective homomorphism.*

Lemma 18.11. *If J is a prime ideal of S, then J^c is a prime ideal of R.*

Proof. S/J is ID, so all subrings of S/J are ID's, so R/J^c is ID, so J^c is prime ideal. $\square$

Recall 18.12. *If R a ring, a subset $S \subseteq R$ is multiplicatively closed iff $1 \in S$ and for all $s, t \in S$, $st \in S$.*
Define a "ring of fractions" $S^{-1}R$.

Theorem 18.13. *Let $X = \{(a, b) : a \in R, b \in S\}$ then*

$$(a_1, b_1) \sim (a_2, b_2) \iff \exists s \in S, s(a_1b_2 - a_2b_1) = 0.$$

Theorem 18.14. *Let the set be $S^{-1}R$. Define the map $\phi : R \to S^{-1}R$ by $\phi : r \mapsto \frac{r}{1}$. Then ϕ is a ring homomorphism but it may not be injective.*

Theorem 18.15. *Universal Property: For all rings A and all homomorphism $\psi : R \to A$ such that $\phi(s)$ is a unit in A for all $s \in S$. There is unique $X : S^{-1}R \to A$ such that $\psi = X \circ \phi$.*

$$\begin{tikzcd} R \arrow[r, "\phi"] \arrow[dr, "\psi"'] & S^{-1}R \arrow[d, dashed, "X"] \\ & A \end{tikzcd}$$

Proof. Define $X\left(\frac{r}{s}\right) = \psi(r)\psi(s)^{-1}$. One checks this is well-defined (since $\psi(s)$ is a unit), is a ring homomorphism, and is the unique map satisfying $\psi = X \circ \phi$. $\square$

19 Lecture 19: Polynomial Rings over UFDs (Feb 25)

19.1 Gauss's Lemma and $R[x]$ as a UFD

Theorem 19.1. *R UFD $\implies$ $R[x]$ UFD*

Remark 19.2. *As R ID for nonzero, $f, g \in R[x]$, $\deg(fg) = \deg(f) + \deg(g)$*

Definition 19.3. *Let $f \in R[x]$ with $f \neq 0$. Then f is **primitive** iff there is no prime $p \in R$ such that $p|f$ in $R[x]$ (i.e. p divides every coefficient of f).*

Remark 19.4. *Informally, the* gcd *of the coefficients of f is a unit.*

Lemma 19.5 (Gauss's Lemma)**.** *If $f, g \in R[x]$ are both primitive, then fg is primitive.*

Proof. Suppose not, and let p be prime in R such that $p \mid fg$ in $R[x]$. Since p is prime, $P = (p)$ is a prime ideal, so R/P is an integral domain. Let $\phi : R \to R/P$ be the quotient map, and define $\phi^* : R[x] \to (R/P)[x]$ by

$$\phi^*\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \phi(a_i) x^i.$$

ϕ^* is a ring homomorphism, so $\phi^*(fg) = \phi^*(f)\phi^*(g)$. But $\phi^*(fg) = 0$ since $p \mid fg$. Since $(R/P)[x]$ is an integral domain, $\phi^*(f) = 0$ or $\phi^*(g) = 0$, i.e. $p \mid f$ or $p \mid g$, contradicting primitivity. □

Lemma 19.6. *Let $f \in R[x]$, $f \neq 0$. Then there are $C \in R$, $g \in R[x]$, g primitive and $f = Cg$*

Proof. $C = \gcd(a_0, ..., a_n)$ if $f(x) = a_0 + a_1 x + \cdots + a_n x^n$. □

Lemma 19.7. *Let $h \in F[x]$, then there is a nonzero $a \in F$ such that $ah \in R[x]$ and ah primitive.*

Lemma 19.8. *Let $f \in R[x]$ be irreducible with $\deg f > 0$. Then f irreducible in $F[x]$*

Proof of Theorem 19.1. **Existence of factorization.** Let $f \in R[x]$ be nonzero and not a unit. Write $f = cg$ where $c \in R$ and $g \in R[x]$ is primitive. If c is a nonunit in R, use that R is a UFD to write $c = r_1 \cdots r_n$ with each r_i irreducible in R; each r_i remains irreducible in $R[x]$.
Next, view $g \in F[x]$. If $\deg(g) > 0$, then g is a nonunit in $F[x]$, and since $F[x]$ is a UFD we can write

$$g = h_1 \cdots h_k \text{ where } h_i \in F[x] \text{ is irreducible in } F[x].$$

Find $a_i \in F^*$ such that $H_i = a_i h_i \in R[x]$ is primitive; then each H_i is also irreducible in $F[x]$. We have

$$\prod_{i=1}^{k} H_i = \left(\prod_{i=1}^{k} a_i\right) g.$$

Write $\prod a_i = A/B$ with $A, B \in R$, $B \neq 0$, so $B \prod_i H_i = Ag$. Since the H_i are primitive, so is $\prod_i H_i$ (Gauss); since g is primitive, comparing contents shows A and B are associates, so $\prod a_i \in R^\times$. WLOG $\prod a_i = 1$, giving $g = \prod H_i$ as a product of irreducibles in $R[x]$.

Uniqueness of factorization. Suppose

$$f = r_1 \cdots r_n \, g_1 \cdots g_s = r'_1 \cdots r'_m \, g'_1 \cdots g'_{s'}$$

with $r_i, r'_j \in R$ irreducible and $g_i, g'_j \in R[x]$ irreducible with $\deg(g_i), \deg(g'_j) > 0$. Since each g_i (resp. g'_j) is primitive, so are $\prod g_i$ and $\prod g'_j$. Equating contents shows $r_1 \cdots r_n$ and $r'_1 \cdots r'_m$ are associates in R; since R is a UFD, $n = m$ and the r_i match the r'_j (up to reorder and units). Each g_i is irreducible in $F[x]$ (Lemma 19.8), and $F[x]$ is a UFD, so $s = s'$ and the g_i match the g'_j up to reorder and units in F^*; combined with primitivity this forces the units to lie in $R^\times$, so they match up to associates in $R[x]$. $\square$

20 Lecture 20: Products and Coproducts in Categories (Mar 2)

20.1 Universal Constructions: Products and Coproducts

Definition 20.1. *Let $\mathcal{C}$ be a category and c, d objects then a cone over c, d is a diagram*

$$\begin{tikzcd} & e \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ c & & d \end{tikzcd}$$

A cone over c, d is a product of c, d iff for all cones

$$\begin{tikzcd} & e' \arrow[dl, "\alpha'"'] \arrow[dr, "\beta'"] & \\ c & & d \end{tikzcd}$$

there is a unique $\phi : e' \to e$ such that the diagram

$$\begin{tikzcd} & e' \arrow[dl, "\alpha'"'] \arrow[d, dashed, "\phi"] \arrow[dr, "\beta'"] & \\ c & e \arrow[l, "\alpha"] \arrow[r, "\beta"'] & d \end{tikzcd}$$

commutes (i.e. $\alpha \circ \phi = \alpha'$ and $\beta \circ \phi = \beta'$).

Example 20.2. *Groups: Let G_1, G_2, H be groups and $\phi : H \to G_1$, $\psi : H \to G_2$ homomorphisms. The product cone is $(G_1 \times G_2, \pi_1, \pi_2)$ where $\pi_1(g_1, g_2) = g_1$ and $\pi_2(g_1, g_2) = g_2$. The unique $\rho : H \to G_1 \times G_2$ with $\rho(h) = (\phi(h), \psi(h))$ makes the diagram*

$$\begin{tikzcd} & H \arrow[dl, "\phi"'] \arrow[d, dashed, "\rho"] \arrow[dr, "\psi"] & \\ G_1 & G_1 \times G_2 \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & G_2 \end{tikzcd}$$

commute ($\pi_1 \circ \rho = \phi$ and $\pi_2 \circ \rho = \psi$).

Recall 20.3. *For any category $\mathcal{D}$, a terminal object is d object such that all objects $\overline{d}$ there is exactly one $\overline{d} \to d$.*

Lemma 20.4. *Given $\mathcal{C}$ and objects c, d make a new category whose objects are the cones over c, d.*

Cone 1

$$\begin{tikzcd} & e \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ c & & d \end{tikzcd}$$

Cone 2

$$\begin{tikzcd} & e' \arrow[dl, "\alpha'"'] \arrow[dr, "\beta'"] & \\ c & & d \end{tikzcd}$$

Drawing the arrow from cone 1 to cone 2 we have

$$\begin{tikzcd} & e \arrow[dl, "\alpha"'] \arrow[d, dashed, "\phi"] \arrow[dr, "\beta"] & \\ c & e' \arrow[l, "\alpha'"'] \arrow[r, "\beta'"'] & d \end{tikzcd}$$

such that $\alpha \circ \phi = \alpha'$ and $\beta \circ \phi = \beta'$.
Being a product of c, d is equivalent to being a terminal object in category of cones over c, d

Definition 20.5. *A co-cone over c, d is a diagram*

$$\begin{tikzcd} c \arrow[dr, "\alpha"'] & & d \arrow[dl, "\beta"] \\ & e & \end{tikzcd}$$

is a coproduct of c, d iff for all cocones

$$\begin{tikzcd} c \arrow[dr, "\alpha'"'] & & d \arrow[dl, "\beta'"] \\ & e' & \end{tikzcd}$$

there is a unique $\phi : e \to e'$ such that the diagram

$$\begin{tikzcd} c \arrow[dr, "\alpha"] \arrow[ddr, "\alpha'"'] & & d \arrow[dl, "\beta"'] \arrow[ddl, "\beta'"] \\ & e \arrow[d, dashed, "\phi"] & \\ & e' & \end{tikzcd}$$

commutes (i.e. $\alpha' \circ \phi = \alpha$ and $\beta' \circ \phi = \beta$).

Example 20.6. *Sets: Let X, Y, Z be sets then define $X \cup Y = (X \times \{0\}) \cup (Y \times \{1\})$. Define $i : x \mapsto (x, 0)$ and $j : y \mapsto (y, 1)$. Then the coproduct is $h(x, 0) = f(x)$ and $h(y, 1) = g(y)$.*

$$\begin{tikzcd} X \arrow[d, "i"] \arrow[dr, "g"] & Y \arrow[d, "f"] \arrow[dl, "j"'] \\ X \cup Y \arrow[r, dashed, "h"] & Z \end{tikzcd}$$

such that $h(i(x)) = f(x)$ and $h(j(y)) = g(y)$.

Definition 20.7. *Let M_1, M_2 be R-modules. Define*

$$M_1 \oplus M_2 = \{(m_1, m_2) : m_i \in M_i\}$$

with componentwise operations.
Coproduct structure. *Define inclusions $i : M_1 \to M_1 \oplus M_2$ by $i(m_1) = (m_1, 0)$ and $j : M_2 \to M_1 \oplus M_2$ by $j(m_2) = (0, m_2)$. For any R-module N and homomorphisms $\psi_1 : M_1 \to N$, $\psi_2 : M_2 \to N$, there is a unique $\rho : M_1 \oplus M_2 \to N$ such that the diagram*

$$\begin{tikzcd}
M_1 \arrow[dr, "i"'] \arrow[drr, bend left, "\psi_1"] & M_2 \arrow[d, "j"] \arrow[drr, shift left, "\psi_2"] & & \\
& M_1 \oplus M_2 \arrow[rr, dashed, "\rho"'] & & N
\end{tikzcd}$$

commutes ($\rho \circ i = \psi_1$ and $\rho \circ j = \psi_2$). Explicitly, $\rho(m_1, m_2) = \psi_1(m_1) + \psi_2(m_2)$.

Product structure. *Define projections $\pi_1 : M_1 \oplus M_2 \to M_1$ by $\pi_1(m_1, m_2) = m_1$ and $\pi_2 : M_1 \oplus M_2 \to M_2$ by $\pi_2(m_1, m_2) = m_2$. For any R-module N and homomorphisms $\psi_1 : N \to M_1$, $\psi_2 : N \to M_2$, there is a unique $\rho : N \to M_1 \oplus M_2$ such that the diagram*

$$\begin{tikzcd}
& N \arrow[dl, "\psi_1"'] \arrow[d, dashed, "\rho"] \arrow[dr, "\psi_2"] & \\
M_1 & M_1 \oplus M_2 \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & M_2
\end{tikzcd}$$

commutes ($\pi_1 \circ \rho = \psi_1$ and $\pi_2 \circ \rho = \psi_2$). Explicitly, $\rho(n) = (\psi_1(n), \psi_2(n))$. Thus for finitely many R-modules, the direct sum $M_1 \oplus M_2$ is simultaneously the product and the coproduct.

21 Lecture 21: Free Modules and Bilinear Maps (Mar 11)

21.1 Free Modules and Universal Properties

Lemma 21.1. *Fix R-module, M and N consider $Hom_R(M, N) = \{\phi : M \to N : \phi R - linear\}$. This forms an R-module.*

Theorem 21.2. *Given ring R and set X, to construct a R-module on X. Let $M = \{g : X \to R, \{x \in X : g(x) \neq 0\}$ is finite$\}$. To make M an R-module, $(g + h)(x) = g(x) + h(x)$ and $(rg)(x) = r(g(x))$.*
We'll write elements of M as $\lambda_1 x_1 + \cdots + \lambda_n x_n$ where $\lambda_i \in R$ and $x_i \in X$

Claim 21.3. *For any R-module N, for any function $f : X \to N$, there is a unique R-linear $\psi : M \to N$ such that $\psi(1x) = f(x)$ for all $x \in X$.*

Proof.

$$\psi\left(\sum_{i=1}^{n} \lambda_i x_i\right) = \sum_{i=1}^{m} \lambda_i f(x_i).$$

□

Theorem 21.4. *Any module is isomorphic to a quotient of a free module.*

Proof. Let N be arbitrary, let $X =$ underlying set of N, $f : X \to N$ where $f(n) = n$. $\psi : M \to N$ with $M = Free_R(X)$, ψ is surjective. □

Remark 21.5. *When I constructed $Free_R(X)$, every element of $Free_R(X)$ is a unique R-linear combination of elements of elements from the set $\{1x : x \in X\}$.*

Definition 21.6. *R-module M is free iff M has a linearly independent generating set.*

Lemma 21.7. *If B is such a set, $M \simeq Free_R(B)$*

21.2 Bilinear Maps and Tensor Motivation

Definition 21.8. *Let M, N, P be R-modules. A function $\phi : M \times N \to P$ is R-bilinear if*

1. *For all n, $m \mapsto \phi(m, n)$ is R-linear*
2. *For all m, $n \mapsto \phi(m, n)$ is R-linear*

Remark 21.9. *Bilinear functions from $M \times N$ to P are very different from linear functions from $M \oplus N$ to P.*

Example 21.10. *Let $R = \mathbb{R}$ and $M = N = \mathbb{R}$:*

1. *The map $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$, $(x, y) \mapsto xy$, is bilinear but not linear.*
2. *The map $\mathbb{R} \oplus \mathbb{R} \to \mathbb{R}$, $(x, y) \mapsto x + y$, is linear but not bilinear.*

Definition 21.11. *Given M, N R-modules we will construct an R-module $M \otimes_R N$ (tensor product of M and N) together with a bilinear map $\otimes : M \times N \to M \otimes_R N$ by $(m, n) \mapsto m \otimes n$ such that for all R-module P and all bilinear $\phi : M \times N \to P$ there is unique R-linear $\rho : M \otimes_R N \to P$ such that $\phi = \rho \circ \otimes$*

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[dr, "\phi"'] & M \otimes N \arrow[d, dashed, "\rho"] \\ & P \end{tikzcd}$$

commutes.
"$\otimes$ is initial in the category of bilinear maps with domain $M \times N$"

Theorem 21.12. *$Hom_R(M, Hom_R(N, P)) \simeq$ bilinear maps from $M \times N \to P \simeq Hom_R(M \otimes N, P)$*

22 Lecture 22: Tensor Product Construction (Mar 11, continued)

22.1 Goal and Preliminaries

Remark 22.1 (Note on Bilinearity)**.** *Let $\phi : M \times N \to P$ be a bilinear map. If $\beta : P \to Q$ is a linear map, then the composition $\beta \circ \phi$ is bilinear.*

Definition 22.2. *(Universal Property of Tensor Products)*
Goal: *Construct a module $M \otimes N$ and a bilinear map $\otimes : M \times N \to M \otimes N$ such that for any module Q and any bilinear map $\psi : M \times N \to Q$, there exists a unique linear map $\beta : M \otimes N \to Q$ that makes the following diagram commute:*

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[dr, "\psi"'] & M \otimes N \arrow[d, dashed, "\exists!\beta"] \\ & Q \end{tikzcd}$$

In other words, $\psi = \beta \circ \otimes$.

Lemma 22.3 (Easy Fact on Quotients)**.** *Let $A \leq B$ be submodules. Then:*

1. *The natural projection $\phi_A : B \to B/A$ is a homomorphism, and it is 0 on A.*
2. *For any module C and homomorphism $\gamma : B \to C$ such that $\gamma(A) = 0$, there is a unique linear map $\delta : B/A \to C$ such that the diagram commutes:*

$$\begin{tikzcd} B \arrow[r, "\phi_A"] \arrow[dr, "\gamma"'] & B/A \arrow[d, dashed, "\exists!\delta"] \\ & C \end{tikzcd}$$

Specifically, δ is defined by mapping $b + A \mapsto \gamma(b)$.

22.2 Construction of the Tensor Product

Definition 22.4 (Constructing $M \otimes N$ and $\otimes$)**.** *Let $X = M \times N$ be the Cartesian product. Construct the free R-module on X, denoted $Fr(X)$, whose elements are finite sums of the form:*

$$\sum \lambda_i(m_i, n_i) \quad \text{where } \lambda_i \in R.$$

Consider the map $\sigma : M \times N \to Fr(X)$ defined by $(m, n) \mapsto 1(m, n)$.
For σ to be bilinear, we would need equations like $(m_1 + m_2, n) - (m_1, n) - (m_2, n) = 0$ and $(\lambda m, n) - \lambda(m, n) = 0$ to hold. However, in $Fr(X)$, these are not *true!*
To fix this, let B be the submodule of $Fr(X)$ generated by all elements of the form that "need to be 0" to satisfy bilinearity. Define the tensor product as the quotient module:

$$M \otimes N = Fr(X)/B$$

and define the map $\otimes : M \times N \to M \otimes N$ by:

$$\otimes : (m, n) \mapsto 1(m, n) + B = m \otimes n.$$

Claim 22.5. *The map $\otimes$ is bilinear.*

Proof. We check one of the bilinearity conditions (the others follow similarly).

$$\begin{aligned}\otimes(\lambda m, n) - \lambda \otimes (m, n) &= 1(\lambda m, n) + B - \lambda(m, n) + B \\ &= [1(\lambda m, n) - \lambda(m, n)] + B \\ &= 0 + B\end{aligned}$$

This evaluates to $0 + B$ because the term in the brackets is one of the generators of the submodule B, meaning it lives inside B. The same logic applies for the other three bilinearity conditions. $\square$

Claim 22.6 (Fulfilling the Goal)**.** *The constructed $M \otimes N$ satisfies the universal property. Let $\psi : M \times N \to Q$ be bilinear. There exists a unique linear $\beta : M \otimes N \to Q$.*

Proof. Let $\phi : Fr(X) \to Q$ be the R-module homomorphism defined on the basis by the unique R-linear extension:

$$\phi\left(\sum \lambda_i(m_i, n_i)\right) = \sum \lambda_i \psi(m_i, n_i) \implies \phi(1(m, n)) = \psi(m, n).$$

We claim $\phi(b) = 0$ for all $b \in B$. Since B is generated by the bilinearity relations, we evaluate ϕ on those generators. For instance:

$$\phi((\lambda m, n) - \lambda(m, n)) = \psi(\lambda m, n) - \lambda\psi(m, n) = 0$$

because ψ is explicitly given as bilinear. Since ϕ maps all 4 types of generators of B to 0, $\phi(B) = 0$.
By our "Easy Fact", there exists a unique linear map $\beta : \frac{Fr(X)}{B} \to Q$ such that the diagram commutes:

$$\begin{tikzcd}
M \times N \arrow[r, "\sigma"] \arrow[dr, "\otimes"'] \arrow[ddr, bend right, "\psi"'] & Fr(X) \arrow[d, "\text{quotient}"] \arrow[dd, bend left, "\phi"] \\
& M \otimes N \arrow[d, dashed, "\beta"] \\
& Q
\end{tikzcd}$$

For elements in $M \otimes N$, we have $m \otimes n = (m, n) + B \implies \beta(m \otimes n) = \psi(m, n)$. $\square$

Remark 22.7 (Uniqueness of $M \otimes N$)**.** *Suppose $\otimes' : M \times N \to M \otimes' N$ is another bilinear map satisfying the universal property. Then, any bilinear map is of the form $\delta \circ \otimes'$ for a linear map δ on $M \otimes' N$. By universal properties, there is a unique isomorphism $I : M \otimes N \to M \otimes' N$ such that $I(m \otimes n) = m \otimes' n$ for all m, n.*

22.3 Tensor Product with the Base Ring

Theorem 22.8. *Let M be an R-module. Then $R \otimes_R M \cong M$.*

Proof. Consider the scalar multiplication map $t : R \times M \to M$ defined by $(r, m) \mapsto rm$. For any bilinear $\psi : R \times M \to Q$, we want to show there exists a unique linear map $\beta : M \to Q$ such that $\psi = \beta \circ t$.

$$\begin{tikzcd} R \times M \arrow[r, "t"] \arrow[dr, "\psi"'] & M \arrow[d, dashed, "\exists!\beta"] \\ & Q \end{tikzcd}$$

If β exists, we must have:

$$\beta(m) = \beta(1 \cdot m) = \beta(t(1, m)) = \psi(1, m).$$

This defines β uniquely. We verify two claims:

1. **Claim 1:** β is linear because ψ is bilinear.
2. **Claim 2:** $\beta \circ t = \psi$.
   Proof: $\psi(r, m) = r\psi(1, m) = \psi(1, rm) = \beta(rm) = \beta(t(r, m))$.

By the uniqueness property of the tensor product, we conclude:

1. $R \otimes M \cong M$.
2. There exists a unique isomorphism $I : R \otimes M \to M$ mapping $r \otimes m \mapsto rm$.

$\square$

23 Lecture 23: Cyclic Modules and PID Structure (Mar 16)

23.1 Cyclic Modules and Direct Summands

Definition 23.1. *An R-module M is cyclic iff there exist m such that $M = Rm$*

Definition 23.2. *M is an R-module, $m \in M$, the annihilator of m is $Ann(m) = \{r : rm = 0\}$*

Lemma 23.3. *For any M R-module any $m \in M$, have R-module homomorhpism $\phi : R \to M$, $\phi : r \mapsto rm$.. Then we have $Im(\phi) = Rm$ and $\ker(\phi) = Ann(m)$ so*

$$Rm \simeq \frac{R}{Ann(m)}.$$

Remark 23.4. *If M cyclic, $M = Rm$, $M \simeq \frac{R}{Ann(m)}$*

Remark 23.5. *If we have I ideal of R and view R-module, then*

$$R/I = R(1 + I) \text{ and } Ann(1 + I) = I.$$

Remark 23.6. *Let R be PID, cyclic R-module $\simeq R/(c)$ for $c \in R$.*

Definition 23.7. *Let M be R-module, P, Q submodules of M are complementary if $M = P + Q$ and $P \cap Q = 0$*

Definition 23.8. *If P, a submodule of M, P is a direct summand of M if there is a complementary Q.*

Example 23.9. *The following are examples of direct summand*

1. *$R = \mathbb{R}, M = \mathbb{R}^2$*
2. *$R = \mathbb{Z}, M = \mathbb{Z}/4\mathbb{Z}$, $P = \{0, 2\}$ then P has no complement*
3. *M_i R-module, $i = 1, ..., n$, $M_1 \oplus \cdots \oplus M_n = \{(m_1, ..., m_n) : m_i \in M_i\}$ with obvious operations*
4. *If $P_1, ..., P_n$ submodules of N and $N = P_1 + P_2 + \cdots + P_n$ and $\sum_{i=1}^n p_i = 0$ ($p_i \in P_i$) then all $p_i = 0, N \simeq P_1 \oplus \cdots \oplus P_n$*

Lemma 23.10. *A free module on a finite set $X = \{x_1, ..., x_n\}$ with $|X| = n$ then $Fr_R(X) \simeq R^n$ by*

$$\sum_{i=1}^n \lambda_i x_i \mapsto (\lambda_1, ..., \lambda_n).$$

23.2 Torsion, Rank, and the Structure Theorem

Definition 23.11. *Let R be ID, M R-module, $m \in M$ is a torsion element if there is $r \neq 0$ such that $rm = 0$.*

Definition 23.12. *R ID, M is R-module, M is torsion free if 0 is the only torsion element.*

Definition 23.13. *Let R be an integral domain and M a torsion-free R-module. Elements $m_1, \ldots, m_n \in M$ are* ***linearly independent*** *if they are distinct and*

$$\sum_{i=1}^{n} r_i m_i = 0 \implies r_1 = r_2 = \cdots = r_n = 0.$$

Definition 23.14. *R ID, M torsion free R-module. M has rank n iff $n = \max\{|I| : I \subseteq M, I$ independent$\}$.*

Theorem 23.15 (Structure Theorem – Invariant Factor Form)**.** *Let R be a PID and let M be a finitely generated R-module. Then M is isomorphic to*

$$\frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)} \oplus R^t,$$

where $t \geq 0$, each $c_i \in R$ is a nonzero nonunit, and $c_i \mid c_{i+1}$ for $1 \leq i < n$. Moreover:

1. *The free rank t is unique.*
2. *The number n of invariant factors is unique.*
3. *The ideals (c_i) are unique; equivalently, the c_i are unique up to associates.*

Lemma 23.16. *If R ID, R^n has rank n.*

Proof. Let $e_i = (0, ..., 1, ..., 0)$ where 1 is in the i-th position. We have $e_1, ..., e_n$ are independent. Let $I \subseteq R^n$ be independent. Let k be the field of fraction of ID R. $I \subseteq R^n \subseteq k^n$.
Claim: I is an independent subset of the k-vs k^n. Prove this by clearing fraction.
Then by linear algebra $|I| \leq \dim_k(k^n) = n$.
The result will follow from the following theorem

Theorem 23.17. *If R PID, N free of rank n and $M \leq N$ then M is free of some rank $m \leq n$.*

There is a basis $z_1, ..., z_n$ of N and there are nonzero nonunit $c_1, ..., c_m$ such that c_i divides c_{i+1} and $c_1 z_1, ..., c_m z_m$ forms a basis of M. □

24 Lecture 24: Submodules of Free Modules over PIDs (Mar 18)

24.1 Diagonalization and Basis Structure

Theorem 24.1. *Let R be a principal ideal domain (PID) and M be a finitely generated R-module. Then*

$$M \simeq \frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)} \oplus R^t$$

where $t \geq 0$, $c_i \neq 0$, $c_i \mid c_{i+1}$ for all $1 \leq i < n$, and the elements c_i are non-units.

Lemma 24.2. *Let R be a ring with $1 \neq 0$, and let R^n be a free R-module of rank n with standard basis $e_i = (0, \ldots, 1, \ldots, 0)$. Then $Hom_R(R^n, R) \simeq R^n$, and it is generated by the dual basis e_j^*, defined by*

$$e_j^* : (\lambda_1, \ldots, \lambda_n) \mapsto \lambda_j.$$

Lemma 24.3. *Let R be a PID, N be a free R-module of rank n, and $M \leq N$ be a submodule. Then:*

1. *M is a free R-module of rank m, where $m \leq n$.*
2. *There exists a basis $z_1, \ldots, z_n$ of N and non-zero elements $d_1, \ldots, d_m \in R$ such that $d_i \mid d_{i+1}$ for all $1 \leq i < m$, and $d_1 z_1, \ldots, d_m z_m$ forms a basis for M.*

Proof. **Step 1: Construction of direct summands.**
We aim to find an element $z \in N$ ($z \neq 0$) and $d \neq 0$ in R such that $dz \in M$, Rz is a direct summand of N, and $R(dz)$ is a direct summand of M.

First, we claim that if $M \neq 0$, there exists $\psi \in \mathrm{Hom}_R(N, R)$ such that $\psi(M) \neq 0$. To show this, without loss of generality, assume $N = R^n$. Let $m = (a_1, \ldots, a_n) \in M$ be a non-zero element. Then $a_j \neq 0$ for some j. Using the dual basis functional, $e_j^*(m) = a_j \neq 0$, which implies $e_j^*(M) \neq 0$.

For any $\phi \in \mathrm{Hom}_R(N, R)$, the image $\phi(M)$ is an ideal of the PID R. Consider the set of ideals:

$$\Sigma = \{\phi(M) : \phi \in \mathrm{Hom}_R(N, R)\}.$$

Since $M \neq 0$, Σ is a non-empty set containing at least one non-zero ideal. Because R is a PID, it is Noetherian, so Σ possesses a maximal element with respect to inclusion. Choose $\psi \in \mathrm{Hom}_R(N, R)$ such that $\psi(M) = (d)$ is maximal in Σ. Choose $y \in M$ such that $\psi(y) = d$.

We claim that $d \mid \phi(y)$ for all $\phi \in \mathrm{Hom}_R(N, R)$. To show this, let $d' = \phi(y)$ and let $e = \gcd(d, d')$. Since R is a PID, we can write $e = \lambda d + \mu d'$ for some $\lambda, \mu \in R$. Then:

$$e = \lambda\psi(y) + \mu\phi(y) = (\lambda\psi + \mu\phi)(y) \in (\lambda\psi + \mu\phi)(M).$$

This implies that:

$$(d) \subseteq (e) \subseteq (\lambda\psi + \mu\phi)(M).$$

By the maximality of $(d) = \psi(M)$ in Σ, we must have $(d) = (e)$. Thus, d and e are associates, which means $d \mid d'$, and consequently $d \mid \phi(y)$.

Now, let $w_1, \ldots, w_n$ be a basis of N, and let $w_j^* \in \mathrm{Hom}_R(N, R)$ be the corresponding dual basis functionals. We can express $y = \sum_{j=1}^n \mu_j w_j$, meaning $w_j^*(y) = \mu_j$. By our previous claim, $d \mid w_j^*(y)$, so $d \mid \mu_j$ for all j. Let:

$$z = \sum_{j=1}^{n} \left(\frac{\mu_j}{d}\right) w_j \in N.$$

Notice that $dz = y$. Furthermore, since $\psi(y) = d$, we have $\psi(dz) = d$. By linearity, $d\psi(z) = d$. Since R is an integral domain and $d \neq 0$, we conclude that $\psi(z) = 1$.

We now claim:

(a) $N = \ker(\psi) \oplus Rz$

(b) $M = (\ker(\psi) \cap M) \oplus Ry$

To show (a), note that if $x \in \ker(\psi) \cap Rz$, then $x = rz$ for some $r \in R$, and $0 = \psi(x) = r\psi(z) = r(1) = r$, so $x = 0$. For any $n \in N$, let $\lambda = \psi(n)$. We can write $n = (n - \lambda z) + \lambda z$. Observe that $\psi(n - \lambda z) = \lambda - \lambda\psi(z) = 0$, so $(n - \lambda z) \in \ker(\psi)$ and $\lambda z \in Rz$. Thus $N = \ker(\psi) \oplus Rz$. The proof of (b) is identical, substituting M for N and y for z.

Step 2: Proof of Part (1).
We proceed by induction on the rank of M (with N fixed). If the rank of M is 0, $M = 0$, which is trivially free. Assume the result holds for submodules of smaller rank. Using our construction, $M = Ry \oplus (\ker(\psi) \cap M)$. Because Ry is generated by a single non-torsion element, it has rank 1. Thus, the rank of $\ker(\psi) \cap M$ is strictly less than the rank of M. By the induction hypothesis, $\ker(\psi) \cap M$ is a free R-module. Adding y to a basis of $\ker(\psi) \cap M$ yields a free basis for M. Thus M is free of rank $m \leq n$.

Step 3: Proof of Part (2).
We proceed by induction on the rank n of N, proving the statement for all submodules $M \leq N$ simultaneously. If $n = 0$, it is trivial. Suppose it holds for free modules of rank $n - 1$. From Step 1, $N = \ker(\psi) \oplus Rz$. Since N has rank n and Rz has rank 1, $\ker(\psi)$ is a free R-module of rank $n - 1$. By the induction hypothesis applied to the submodule $\ker(\psi) \cap M \leq \ker(\psi)$, there exists a basis $z_2, \ldots, z_n$ of $\ker(\psi)$ and elements $d_2, \ldots, d_m \in R$ with $d_2 \mid d_3 \mid \cdots \mid d_m$ such that $d_2 z_2, \ldots, d_m z_m$ is a basis for $\ker(\psi) \cap M$.

Let $z_1 = z$ and $d_1 = d$. Then $z_1, z_2, \ldots, z_n$ is a basis for N, and $d_1 z_1, d_2 z_2, \ldots, d_m z_m$ is a basis for M.

It remains only to show that $d_1 \mid d_2$. Define a homomorphism $f : N \to R$ by mapping $z_1 \mapsto 1$, $z_2 \mapsto 1$, and $z_i \mapsto 0$ for $i > 2$. We evaluate $f(M)$: $f(d_1 z_1) = d_1$ and $f(d_2 z_2) = d_2$, so the ideal $f(M)$ contains both d_1 and d_2, meaning $f(M) = (d_1, d_2)$. Because $(d_1) = \psi(M)$ is maximal among such ideals, and $(d_1) \subseteq (d_1, d_2) = f(M)$, we must have $(d_1) = (d_1, d_2)$. This implies $d_2 \in (d_1)$, so $d_1 \mid d_2$. This completes the proof. $\square$

25 Lecture 25: Structure Theorem and Adjoint Functors (Mar 20)

25.1 Decomposition of Finitely Generated Modules

Theorem 25.1. *Let F be a finitely generated module over a PID R, say $F = Rf_1 + \cdots + Rf_t$ with $f_i \in F$. Define $\pi : R^t \to F$ by $\pi : (\lambda_1, \ldots, \lambda_t) \mapsto \sum_{i=1}^t \lambda_i f_i$. Then π is a surjective homomorphism, so*

$$F \simeq \frac{R^t}{\ker(\pi)}.$$

Setting $N = R^t$ (of rank t) and $M = \ker(\pi)$ (a submodule of N, hence free of some rank $s \leq t$ by the submodule theorem), we get $F \simeq N/M$.

Theorem 25.2. *Let R be a PID, M be a finitely generated R-module. Then M is isomorphic to*

$$\frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)} \oplus R^t$$

where $t \geq 0$, $c_i \neq 0$, $c_i \mid c_{i+1}$ for all $1 \leq i < n$, and the elements c_i are non-units.

Lemma 25.3. *By previous result, we get a basis $n_1, ..., n_t$ for N and nonzero constants $r_1, ..., r_s \in R$ such that $r_i | r_{i+1}$ for $1 \leq i < s$ and $r_1 n_1, ..., r_s n_s$ is a basis for M.*

Exercise 25.4.

$$\frac{N}{M} \simeq \frac{R}{(r_1)} \oplus \cdots \oplus \frac{R}{(r_s)} \oplus R^{t-s}.$$

Lemma 25.5. *If r nonzero nonunit, $r = u\pi_1^{k_1} \cdots \pi_n^{k_n}$ for u unit, π non associate and irreducibles.*

Theorem 25.6 (Chinese Remainder Theorem)**.** *If $(r) = (\pi_1)^{k_1} \cdots (\pi_n)^{k_n}$ then*

$$\frac{R}{(r)} \simeq \frac{R}{(\pi_1^{k_1})} \oplus \cdots \oplus \frac{R}{(\pi_n^{k_n})}.$$

25.2 Adjoint Functors

Definition 25.7. *Let $\mathcal{U}$ be "forgetful" functor from group to set.*

$$\mathcal{U}(G) = \text{underlying set of } G.$$

If $\phi : G \to H$, $\mathcal{U}(\phi)$ is ϕ viewed as a function.

Lemma 25.8. *We have for any group G and any set X, there is a bijection between $Hom_{Gr}(Fr(X), G)$ and $Hom_{Set}(X, \mathcal{U}(G))$*

Definition 25.9 (Natural Transformation)**.** *Let $F_1, F_2 : \mathcal{C} \to \mathcal{D}$. A natural transformation from $F_1 \to F_2$ is a family $v = (v_c)$, for c object of $\mathcal{C}$, $F_1 c \xrightarrow{v_c} F_2 c$ in $\mathcal{D}$ such that for all $c \xrightarrow{\alpha} c'$*

$$\begin{tikzcd} F_1 c \arrow[r, "v_c"] \arrow[d, "F_1\alpha"'] & F_2 c \arrow[d, "F_2\alpha"] \\ F_1 c' \arrow[r, "v_{c'}"'] & F_2 c' \end{tikzcd}$$

commutes.

Definition 25.10 (Opposite Category)**.** *$\mathcal{C}$ category then $\mathcal{C}^{op}$ is the category with the same objects but with the arrows reversed.*

26 Lecture 26: Adjunctions and Tensor–Hom (Mar 23)

26.1 Adjunction Definitions and Naturality

Remark 26.1. *If $Fc \xrightarrow{\alpha} d$ (in $\mathcal{D}$) write α^* for $v_{cd}(\alpha)$ and if $c \xrightarrow{\beta} Gd$ (in $\mathcal{C}$) write β_* for $v_{cd}^{-1}(\beta)$.*

Definition 26.2. *Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ with $\mathcal{C}, \mathcal{D}$ categories and F, G functors. An adjunction between F and G is a family of bijections $v_{cd} : Hom_{\mathcal{D}}(Fc, d) \simeq Hom_{\mathcal{C}}(c, Gd)$ for objects c of $\mathcal{C}, d$ of $\mathcal{D}$ such that*

1. *Naturality in c: If $y = \delta(F\alpha)$ commutes then $y^* = \delta^*\alpha$ commutes.*

$$\begin{tikzcd} Fc_1 \arrow[r, "F\alpha"] \arrow[dr, "y"'] & Fc_2 \arrow[d, "\delta"] \\ & d \end{tikzcd} \implies \begin{tikzcd} c_1 \arrow[r, "\alpha"] \arrow[dr, "y^*"'] & c_2 \arrow[d, "\delta^*"] \\ & Gd \end{tikzcd}$$

2. *Naturality in d: Let $d_1 \xrightarrow{\beta} d_2$*

Recall 26.3. *Universal Property of Products:*

Theorem 26.4. *If G is a right adjoint of some F then G preserves products. If $d \xrightarrow{\pi_1} d_1$ and $d \xrightarrow{\pi_2} d_2$ exhibits $d = d_1 \times d_2$ in $\mathcal{D}$, then*

$$\begin{tikzcd} & d \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ d_1 & & d_2 \end{tikzcd} \implies \begin{tikzcd} & Gd \arrow[dl, "G\pi_1"'] \arrow[dr, "G\pi_2"] & \\ Gd_1 & & Gd_2 \end{tikzcd}$$

and Gd is a product of Gd_1 and Gd_2 in $\mathcal{C}$ (so $Gd \simeq Gd_1 \times Gd_2$).

Remark 26.5. *Let R be a ring, M, N, P be R-modules. We can view $\phi : M \times N \to P$ bilinear as an element of $Hom(M, Hom(N, P))$. This is from $\phi(m, -)$ is a linear map from N to P and $m \mapsto \phi(m, -)$ is a linear map from M to $Hom(N, P)$.*

Lemma 26.6. *$Hom(M \otimes N, P) \simeq Hom(M, Hom(N, P)) \simeq$ bilinear maps from $M \times N \to P$*

Lemma 26.7. *Fixing N, this gives adjunction between $- \otimes N$ and $Hom(N, -)$*

27 Lecture 27: Noetherian Modules and Integrality (Mar 25)

27.1 Noetherian Conditions and Closure Properties

Recall 27.1. *If R PID and M finitely generated R-module then*

$$M \simeq \frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)} \oplus R^t.$$

where c_i nonzero nonunit and $c_i | c_{i+1}$.

Corollary 27.2. *If R PID and M finitely generated torsion-free then M is free.*

Recall 27.3. *An R-module M is noetherian if it satisfies any of the equivalent condition*

1. *Every $N \leq M$ is finitely generated*
2. *Every chain $N_0 \subseteq N_1 \subseteq \cdots$ of submodules is eventually constant.*
3. *Any nonempty set of submodules of M has a maximal element*

Lemma 27.4. *If M noetherian and $N \leq M$ then*

1. *N noetherian*
2. *M/N noetherian (submodules of $M/N \simeq$ submodules M that contain N)*

Lemma 27.5. *If $N \leq M$ and both N and M/N are noetherian then M noetherian.*

Proof. Let $(K_i)_i$ with $K_0 \subseteq K_1 \subseteq \cdots$ and K_i submodules of M. Then $(K_i \cap N)_i$ stabilizes. Consider $(\phi_N[K_i])_i$ with $\phi_N : M \to M/N$ stablizes. Let j be so large that $i \geq j$, $K_i \cap N = K_j \cap N$ and $\frac{K_i+N}{N} = \frac{K_j \cap N}{N}$.
Let $i \geq j$, we've already shown $K_j \subseteq K_i$ so we just have to show that $K_i \subseteq K_j$. Let $k \in K_i$ then $k + N \in \frac{K_i+N}{N} = \frac{K_j+N}{N}$. So there is $k' \in K_j$ such that $k + N = k' + N$ and so $k - k' \in N$. As $k, k' \in K_i$ we also have $k - k' \in N \cap K_i = N \cap K_j$. □

Lemma 27.6. *If M_1, M_2 are noetherian R-modules then $M_1 \oplus M_2$ is noetherian.*

Proof. Note $M_1 \simeq M_1' = M_1 \oplus 0 \leq M_1 \oplus M_2$, and $(M_1 \oplus M_2)/M_1' \simeq M_2$. Both M_1' and M_2 are Noetherian, so by the previous lemma $M_1 \oplus M_2$ is Noetherian. □

Lemma 27.7. *A finite direct sum of noetherian R-modules is noetherian.*

Proof. By induction. □

Lemma 27.8. *If R is noetherian ring then for all t, R^t is a noetherian R-module.*

Proof. By the previous lemma □

Lemma 27.9. *If $S \subseteq R$ MC and R noetherian ring then $S^{-1}R$ is a noetherian ring.*

Lemma 27.10. *If R is noetherian then all finitely generated R-modules are noetherian.*

Proof. Let M be $Rm_1 + \cdots + Rm_t$. Let $\phi : R^t \to M$ by $\phi(\lambda_1, ..., \lambda_t) = \sum_{i=1}^{t} \lambda_i m_i$. We have ϕ is surjective so $M \simeq \frac{R^t}{\ker(\phi)}$. By previous theorem, M is noetherian. □

Definition 27.11. *An R-module M is faithful if for all $r \in R$, if $rm = 0$ for all $m \in M$ then $r = 0$*

Remark 27.12. *If $R \leq S$, then*

1. *S can be viewed as an R-module.*
2. *S is faithful as an R-module: if $rs = 0$ for all $s \in S$, then in particular $r \cdot 1_S = 0$, so $r = 0$.*

Lemma 27.13. *Suppose $R \leq S$ and $a \in S$. The least subring of S containing $R \cup \{a\}$ equals $\{f(a) : f \in R[x]\}$; we denote it $R[a]$.*

Theorem 27.14. *The evaluation map $\phi : R[x] \to S$, $\phi(f) = f(a)$, is a ring homomorphism with image $R[a]$, so*

$$R[a] \simeq \frac{R[x]}{\ker(\phi)}.$$

27.2 Integral Elements over Subrings

Definition 27.15. *Let $R \leq S$, let $a \in S$ with a integral over R if there is monic $f \in R[x], f(a) = 0$.*

28 Lecture 28: Integrality and Flat Modules (Mar 27)

28.1 Integrality Criteria and Consequences

Recall 28.1. *For R, S where $R \leq S$ then for $s \in S$, $R[s] = \{r_0 + r_1 s + r_2 s^2 + \cdots + r_k s^k\}$.*

Definition 28.2. *If $R \leq S$, $s \in S$ integral over R if $f(s) = 0$ for some monic $f \in R[x]$.*

Lemma 28.3. *The following are equivalent where $R \leq S, s \in S$*

1. *s integral over R*
2. *$R[s]$ is finitely generated as R-module.*
3. *There is a faithful $R[s]$-module which is finitely generated as an R-module.*

Proof. (1) $\implies$ (2): There exist $n \geq 1$ and $c_0, \ldots, c_{n-1} \in R$ such that $s^n = \sum_{i<n} c_i s^i$. By induction, for all $m \geq n$, s^m lies in the R-span of $\{s^i : i < n\}$. Hence $R[s] = \sum_{i<n} Rs^i$ is finitely generated as an R-module.
(2) $\implies$ (3): Take $M = R[s]$. It is finitely generated as an R-module, and it is a faithful $R[s]$-module (since $1 \in R[s]$, if $r \in R[s]$ annihilates all of $R[s]$, then $r \cdot 1 = r = 0$). □

Theorem 28.4 (Cayley–Hamilton)**.** *Let A be an $n \times n$ matrix over a commutative ring and let $p(\lambda) = \det(\lambda I_n - A) = \lambda^n + \sum_{i<n} r_i \lambda^i$ be its characteristic polynomial. Then $p(A) = 0$.*

Proof. To show (3) $\implies$ (1), let M be faithful $R[s]$-module with generating set $m_1, ..., m_n$. For each j, $sm_j = \sum_{i=1}^n a_{i,j} m_i$ for some $a_{i,j} \in R$. Let $A = (a_{i,j})$ and $p(\lambda) = \det(\lambda I - A) \in R[\lambda]$. By Cayley-Hamilton, $p(A) = 0$, verify $p(s)m_j = 0$ for all j. As m_j's generate M, $p(s)M = 0$ for all $m \in M$. As M faithful, $p(s) = 0$ □

Remark 28.5. *To motivate this, $\mathbb{Z} \leq \mathbb{C}$ and we say $\alpha \in \mathbb{C}$ is a algebraic integer iff α integral over $\mathbb{Z}$.*

Theorem 28.6. *$\{s \in S : s$ integral over $R\}$ is a subring of S.*

Proof. Let s, t both be integral over R. I claim that all $u \in R[s,t]$ are integral over R. In particular, $s + t, st$ are integral over R. We have $R[s], R[t]$ both finitely generated R-modules, say $R[s] = R$-span of $\{a_1, ..., a_k\}$ and $R[t] = R$-span of $\{b_1, ..., b_l\}$. Let $R[s,t] = R$-span of $\{a_i b_j | i \leq k, j \leq l\}$. Now $R[s,t]$ is finitely generated R-module and faithful. □

28.2 Flat Modules

Recall 28.7. *View* $- \otimes N$ *as a functor from* M *to* $M \otimes N$*. If* $M_1 \xrightarrow{\alpha} M_2$ *then* $M_1 \otimes N \xrightarrow{\alpha \otimes N} M_2 \otimes N$ *by* $\alpha \otimes N \left(\sum_{i=1}^{n} m_i \otimes n_i\right) = \sum_{i=1}^{s} \alpha m_i \otimes n_i$*.*

Theorem 28.8. *From the homework*

$$A \to B \to C \to 0.$$

exact then

$$A \otimes N \to B \otimes N \to C \otimes N \to 0.$$

exact.

Definition 28.9. *An* R*-module* N *is flat if, for all exact sequences* $A \to B \to C$*, the induced sequence* $A \otimes_R N \to B \otimes_R N \to C \otimes_R N$ *is exact.*

Theorem 28.10. *TFAE:*

1. N *flat*
2. *For all* $0 \to A \to B$ *exact,* $0 \to A \otimes N \to B \otimes N$ *exact.*
3. *For all ideal* I *of* R *if* $I \overset{i}{\hookrightarrow} R$ *then* $I \otimes N \xrightarrow{i \otimes N} R \otimes N$ *injective.*
4. (3) *but only for finitely generated* I*.*
5. *(Equational criterion for flatness) Whenever* $\sum_{i=1}^{n} r_i n_i = 0$ *with* $r_i \in R$ *and* $n_i \in N$*, there exist* $\overline{n_1}, \ldots, \overline{n_t} \in N$ *and* $\lambda_{i,j} \in R$ *such that* $n_i = \sum_{j=1}^{t} \lambda_{i,j} \overline{n_j}$ *for each* i*, and* $\sum_{i=1}^{n} r_i \lambda_{i,j} = 0$ *for each* j*.*

29 Lecture 29: Projective Modules

29.1 Module Extensions and Projective Modules

Proposition 29.1. *Given R-modules A and C, the "extension problem" asks us to find a module B where $A \leq B$ and $B/A \simeq C$. This is equivalent to fitting B into the short exact sequence*

$$0 \to A \to B \to C \to 0.$$

A trivial answer is always $B = A \oplus C$, since $A \simeq A \oplus 0$, giving

$$\frac{A \oplus C}{A \oplus 0} \simeq C.$$

Example 29.2. *Let $R = \mathbb{Z}$ and $A = C = C_2$ (the cyclic group of order 2). Then B can be the trivial extension $C_2 \oplus C_2$, or the non-trivial extension $B = C_4$.*

Definition 29.3. *Let $0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \to 0$ be a short exact sequence of R-modules. This sequence **splits** if and only if there is a homomorphism $\gamma : C \to B$ such that $\beta \circ \gamma = 1_C$.*

Example 29.4. *Consider the trivial exact sequence $0 \to A \xrightarrow{\alpha} A \oplus C \xrightarrow{\beta} C \to 0$ with $\alpha : a \mapsto (a, 0)$ and $\beta : (a, c) \mapsto c$. Then the map $\gamma : C \to A \oplus C$ given by $\gamma : c \mapsto (0, c)$ splits the sequence. Up to isomorphism, this is the only way a sequence splits.*

Lemma 29.5. *More precisely, if $\gamma : C \to B$ splits the sequence $0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \to 0$, then $\mathrm{im}(\alpha)$ and $\mathrm{im}(\gamma)$ are complementary submodules in B, with $\mathrm{im}(\alpha) \simeq A$ and $\mathrm{im}(\gamma) \simeq C$. Thus, $B \simeq A \oplus C$.*

Proof. By exactness, $\mathrm{im}(\alpha) = \ker(\beta)$. Because $\beta \circ \gamma = 1_C$, we have $\mathrm{im}(\alpha) \cap \mathrm{im}(\gamma) = \ker(\beta) \cap \mathrm{im}(\gamma) = 0$.
Given any $b \in B$, we can write it as $b = (b - \gamma\beta b) + \underbrace{\gamma\beta b}_{\in \mathrm{im}(\gamma)}$. The first term is in $\mathrm{im}(\alpha)$ because applying β gives:

$$\beta(b - \gamma\beta b) = \beta b - \beta\gamma\beta b = \beta b - 1_C(\beta b) = 0.$$

Therefore, $B = \mathrm{im}(\alpha) \oplus \mathrm{im}(\gamma) \simeq A \oplus C$. □

Definition 29.6. *An R-module M is **projective** if and only if for every surjective homomorphism $\pi : B \twoheadrightarrow C$ and every homomorphism $\alpha : M \to C$, there exists a homomorphism $g : M \to B$ such that $\pi \circ g = \alpha$. Equivalently, the following diagram commutes:*

$$\begin{tikzcd} & M \arrow[dl, dashed, "g"'] \arrow[d, "\alpha"] \\ B \arrow[r, two heads, "\pi"] & C \end{tikzcd}$$

Lemma 29.7. *The following are equivalent (TFAE) for an R-module M:*

1. *M is projective.*
2. *The functor $\mathrm{Hom}_R(M, -)$ is exact. (i.e., for any short exact sequence $0 \to A \to B \to C \to 0$, the induced sequence $0 \to \mathrm{Hom}(M, A) \to \mathrm{Hom}(M, B) \to \mathrm{Hom}(M, C) \to 0$ is exact).*
3. *Every short exact sequence $0 \to K \to N \to M \to 0$ ending in M splits.*
4. *There exists an R-module N such that $M \oplus N$ is a free module.*

Proof. (1) $\implies$ (4): Let X be the underlying set of M and let $F = \mathrm{Free}_R(X)$ with the natural surjection $\pi : F \twoheadrightarrow M$. Applying (1) to π and $1_M : M \to M$, there exists $i : M \to F$ such that $\pi \circ i = 1_M$:

$$\begin{tikzcd} M \arrow[rr, "1_M"] \arrow[dr, "i"'] & & M \\ & F \arrow[ur, two heads, "\pi"'] & \end{tikzcd}$$

Let $N = \ker(\pi)$. By the splitting lemma (Lemma 1), $i(M)$ and N are complementary in F. Since $i(M) \simeq M$, we get $F \simeq i(M) \oplus N \simeq M \oplus N$.
(4) $\implies$ (1): Suppose $M \oplus N$ is free. We want to show M is projective. Suppose we are given $B \xrightarrow{\pi} C \to 0$ and $\alpha : M \to C$.

$$\begin{tikzcd} & M \oplus N \arrow[dl, dashed, "\beta"'] \arrow[d, "\overline{\alpha}"] \\ B \arrow[r, two heads, "\pi"] & C \end{tikzcd}$$

Define $\overline{\alpha} : M \oplus N \to C$ by $\overline{\alpha}(m, n) = \alpha(m)$. Let X be a basis of the free module $M \oplus N$. For each $x \in X$, the map π is surjective, so we can choose $f(x) \in B$ such that $\pi(f(x)) = \overline{\alpha}(x)$.

Because $M \oplus N$ is free on X, there is a unique homomorphism $\beta : M \oplus N \to B$ extending f, meaning $\beta(x) = f(x)$ for all $x \in X$. By choice of f, $\pi\beta(x) = \overline{\alpha}(x)$ for all basis elements. Since X generates $M \oplus N$, we have $\pi \circ \beta = \overline{\alpha}$.

Finally, define $g : M \to B$ by $g(m) = \beta(m, 0)$. Then for any $m \in M$:

$$\pi(g(m)) = \pi(\beta(m, 0)) = \overline{\alpha}(m, 0) = \alpha(m).$$

Thus, g is the required lift, proving M is projective. $\square$

30 Lecture 30: Field Theory

Recall 30.1. *Basic properties of fields:*

1. $0 \neq 1$ *and all nonzero elements are units.*
2. *Ring with exactly* 2 *ideals,* (1) *and* (0).
3. *Field of fraction of an ID is a field*
4. *Subring of a field is ID*
5. R/I *field iff* I *maximal ideal*

Lemma 30.2. *If* k, l *both fields and* $\phi : k \to l$ *homomorphism then* $\phi(1) = 1 \neq 0$ *and as* $\ker(\phi)$ *is an ideal then* $\ker(\phi) \neq k \implies \ker(\phi) = 0 \implies \phi$ *injective. So* $\phi[k] \simeq k$.

Remark 30.3. *Terminology: If* k *subfield of* l *then* l *is an extension of* k *(often written as* l/k*)*

Definition 30.4. *Let* l/k *be a field extension. We define:*

1. $[l : k] =$ *the dimension of* l *as a* k*-vector space (the* degree *of the extension).*
2. *For* $\alpha \in l$, α *is* ***algebraic over*** k *if* $f(\alpha) = 0$ *for some nonzero* $f \in k[x]$*; otherwise* α *is* ***transcendental over*** k.
3. *For* $\alpha \in l$, $k(\alpha)$ *is the least subfield of* l *containing* $k \cup \{\alpha\}$. *More generally, if* $A \subseteq l$, *then* $k(A)$ *is the least subfield of* l *containing* $k \cup A$.

Lemma 30.5. $k(\alpha) = \left\{ \frac{f(\alpha)}{g(\alpha)} : f, g \in k[x], g(\alpha) \neq 0 \right\}$

Definition 30.6. $k(x)$ *is the field of fractions of* $k[x]$.

Lemma 30.7. $k(x) \simeq k(\alpha)$ *so* $\frac{f(\alpha)}{g(\alpha)}$ *corresponds* $\frac{f}{g}$ *when* α *is transcendental.*

Proposition 30.8. *If we analyize $k(\alpha)$ depending on if α is algebraic or transcendental then we have the following:*

1. *If α is transcendental over k, consider $e_\alpha : k[x] \to l$ by $e_\alpha : f \mapsto f(\alpha)$ and as e_α homomorphism and α transcendental, $\ker(e_\alpha) = 0$ so $im(e_\alpha) \simeq k[x]$. So $f(\alpha)$ corresponds with f.*
2. *If α is algebraic over k, consider same homomorphism, $im(e_\alpha) = k[\alpha]$. Since α is algebraic, $\ker(e_\alpha) \neq 0$; by the First Isomorphism Theorem,*

$$\operatorname{im}(e_\alpha) \simeq \frac{k[x]}{\ker(e_\alpha)}.$$

Since $k[\alpha]$ is a subring of the field l, it is an integral domain, so $\ker(e_\alpha)$ is a nonzero prime ideal of the PID $k[x]$. Therefore:

   (a) *$\ker(e_\alpha) = (f)$ for a unique monic $f \in k[x]$.*
   (b) *f is prime and hence irreducible.*
   (c) *(f) is a maximal ideal.*
   (d) *$k[x]/(f)$ is a field, so $k[\alpha] = k(\alpha)$.*
   (e) *Let m_α (the* minimal polynomial of α over k*) be the unique monic generator $\ker(e_\alpha) = (m_\alpha)$, and let $n = \deg(m_\alpha)$. Then $k(\alpha) \simeq k[x]/(m_\alpha)$, with $f(\alpha)$ corresponding to $f + (m_\alpha)$. For any $f \in k[x]$, division gives a unique representation $f = m_\alpha q + r$ with $r = \sum_{i<n} c_i x^i$, $c_i \in k$; so $f + (m_\alpha) = r + (m_\alpha)$ with r unique.*
   (f) *$1, \alpha, \ldots, \alpha^{n-1}$ form a basis for $k(\alpha)$ as a k-vector space; in particular $[k(\alpha) : k] = n = \deg(m_\alpha)$.*

Theorem 30.9 (Tower Law)**.** *Let k_1, k_2, k_3 be fields with $k_1 \subseteq k_2 \subseteq k_3$. Let $B \subseteq k_3$ be a basis for k_3 as a k_2-vector space, and $A \subseteq k_2$ a basis for k_2 as a k_1-vector space. Then the products $AB = \{ab : a \in A, b \in B\}$ are pairwise distinct (i.e. $a_1b_1 = a_2b_2 \iff a_1 = a_2$ and $b_1 = b_2$), and form a basis for k_3 as a k_1-vector space. In particular, $[k_3 : k_1] = [k_3 : k_2] \cdot [k_2 : k_1]$.*

Proof. **Spanning.** Let $c \in k_3$. Since B spans k_3 over k_2, write

$$c = \sum_{i=1}^{t} \lambda_i b_i, \quad \lambda_i \in k_2,\ b_i \in B.$$

Since A spans k_2 over k_1, we can further decompose

$$\lambda_i = \sum_{j=1}^{s_i} \mu_{i,j} a_{i,j}, \quad \mu_{i,j} \in k_1,\ a_{i,j} \in A.$$

Substituting, $c = \sum_{i,j} \mu_{i,j}(a_{i,j}b_i)$ lies in the k_1-span of AB.
Linear independence. Suppose $\sum_{a\in A,\, b\in B} \mu_{a,b}(ab) = 0$ with $\mu_{a,b} \in k_1$ (finitely many nonzero). Group by b:

$$\sum_{b\in B} \underbrace{\left(\sum_{a\in A} \mu_{a,b}\, a\right)}_{\in k_2} \cdot b = 0.$$

Since B is k_2-linearly independent, each inner sum $\sum_a \mu_{a,b} a = 0$ in k_2. Since A is k_1-linearly independent, each $\mu_{a,b} = 0$.
In particular, the multiplication map $A \times B \to AB$ is injective, so $|AB| = |A||B|$. $\square$

Remark 30.10. *We have $[k(x) : k]$ is infinite so if $[l : k] < \infty$ then for all $\alpha \in l$, $k \leq k(\alpha) \leq l$, $[k(\alpha) : k] < \infty$. So α is algebraic over k.*

31 Lecture 31: Field Extensions

Lemma 31.1. *Let $k \leq l$. If $[l : k] < \infty$, then $[k(\alpha) : k] < \infty$ for all $\alpha \in l$ so α is algebraic over k.*

Lemma 31.2. *If l/k, $\alpha_1, ..., \alpha_n \in l$ and α_i algebraic over k then $[k(\alpha_1, ..., \alpha_n) : k] < \infty$.*

Proof. By induction for $n = 2$, α_1, α_2 algebraic over k implies α_2 is algebraic over $k(\alpha_1)$. So $[k(\alpha_1, \alpha_2) : k(\alpha_1)] \leq [k(\alpha_2) : k]$. So by induction hypothesis,

$$[k(\alpha_1, ..., \alpha_n) : k] \leq \prod_{i=1}^{n} [k(\alpha_i) : k] < \infty..$$

□

Proposition 31.3. *Adding Roots: Let $f \in k[x]$ be monic and irreducible, then (f) is maximal ideal in $k[x]$, so $\frac{k[x]}{(f)}$ is a field. We have $c \in k \mapsto c + (f)$ is injective. Let $l = \frac{k[x]}{(f)}$ and identify $c \in k$ with $c + (f)$. Then $k \leq l$ and if we let $\alpha = x + (f)$ then $l = k(\alpha), f = m_\alpha$ and $[k(\alpha) : k] = \deg(f)$.*

Example 31.4. *Let $k = \mathbb{R}$ and $f = x^2 + 1$ irreducible and monic. We can form $\frac{\mathbb{R}[x]}{(x^2+1)} = \{a + bx + (x^2 + 1) : a, b \in \mathbb{R}[x]\}$. We can identify $a \in \mathbb{R}$ with $a + (x^2 + 1)$. Let $i = x + (x^2 + 1)$. So now every element is $a + bi$ for unique $a, b \in \mathbb{R}$.*
Say $i^2 = x^2 + (x^2 + 1) = -1 + (x^2 + 1)$.

Definition 31.5. *Let k be a field and $f \in k[x]$ splits iff f is a product of linear factors in $k[x]$.*
Equivalently, f splits iff

$$f = c \prod_{i=1}^{n} (x - \alpha_i).$$

Definition 31.6. *Let k be a field and $f \in k[x]$, l/k is a splitting field extension iff f splits in l and $l = k(\alpha_1, ..., \alpha_n)$ for $\alpha_i \in l$ algebraic over k.*

Example 31.7. *Splitting Fields:*
1. *$\mathbb{C}/\mathbb{R}$ splits for $x^2 + 1$ so $\mathbb{C} = \mathbb{R}(i, -i)$*
2. *$\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ splits for $x^2 - 2$.*

Theorem 31.8. *Let $f \in k[x]$ such that $f \neq 0, \deg(f) = n$ then there is a splitting field extension l/k for f with $[l : k] \leq n!$.*

Proof. We'll show this by induction:
(Base Case) $n = 0, l = k$
(Inductive Step) Let $g \in k[x]$ be an irreducible factor of f. Let $k' = k(\alpha)$ where $g(\alpha) = 0$,

constructed as before. $[k(\alpha) : k] = \deg(g) \leq \deg(f) \leq n$. As $g|f$, $f(\alpha) = 0$. In $k'[x]$, since $f(\alpha) = 0, x - \alpha | f$, say $f = (x - \alpha)f'$ with $f' \in k'[x]$ and $\deg(f') = n - 1$. By IH for $n - 1$, k', f', set l/k', f' splits in $l[x]$. We have $l = k'(B)$ so B is root of f' in l and $[l : k'] \leq (n - 1)!$. We have to verify that l/k is sfe for f and $[l : k] \leq n!$. $\square$

32 Lecture 32: Splitting Field Extension

Lemma 32.1. *Let l/k, $\alpha \in l$, let $f \in k[x]$ monic, irreducible and $f(\alpha) = 0$ then $f =$ minimal polynomial of α over k.*

Proof. As $f(\alpha) = 0, \alpha$ is algebraic over k. Let $m =$ minimal polynomial of α over k. Then as $f(\alpha) = 0$, $m|f$ in $k[x]$. So m, f associates and monic so $m = f$. □

Lemma 32.2. *Let $l_1/k_1, l_2/k_2$ be field extensions and $\alpha_1 \in l_1, \alpha_2 \in l_2$ with m_i the minimal polynomial of α_i over k_i. Let $\rho : k_1 \to k_2$ be a field isomorphism such that $\rho_*(m_1) = m_2$ (where ρ_* extends ρ coefficientwise to $k_1[x] \to k_2[x]$). Then there is a unique field isomorphism $\tau : k_1(\alpha_1) \to k_2(\alpha_2)$ such that $\tau|_{k_1} = \rho$ and $\tau(\alpha_1) = \alpha_2$.*

Proof. Let $n = \deg(m_1) = \deg(m_2)$, if τ exists

$$\tau\left(\sum_{i=0}^{n-1} c_i\alpha_1^i\right) = \sum_{i=0}^{n-1} \rho(c_i)\alpha_2^i.$$

So τ is unique.
We have

$$k_1(\alpha_1) \simeq \frac{k_1[x]}{(m_1)} \simeq \frac{k_2[x]}{(m_2)} \simeq k_2(\alpha_2).$$

by $f(\alpha_1) \leftrightarrow f + (m_1) \leftrightarrow \rho(f) + (m_2) \leftrightarrow \rho(f)(\alpha_2)$. So τ exists. □

Theorem 32.3 (Uniqueness of Splitting Field Extensions)**.** *If $\rho : k_1 \to k_2$ is a field isomorphism, $f_1 \in k_1[x]$, $\rho_*(f_1) = f_2 \in k_2[x]$, and l_i/k_i is a splitting field extension for f_i $(i = 1, 2)$, then there is an isomorphism $\tau : l_1 \to l_2$ extending ρ, i.e. $\tau|_{k_1} = \rho$.*

Proof. By induction on $[l_1 : k_1]$.
(Base Case) $[l_1 : k_1] = 1 \implies l_1 = k_1$, so f_1 splits completely in $k_1[x]$, and similarly for f_2 in $k_2[x]$ as ρ is an isomorphism. So $l_2 = k_2$, and we can simply take $\tau = \rho$.
(Induction Step) Let $[l_1 : k_1] = n > 1$. Since l_1 is the splitting field extension of k_1 for f_1, there must be a root α_1 of f_1 with $\alpha_1 \in l_1 \setminus k_1$. Let $\overline{n} = [k_1(\alpha_1) : k_1] > 1$ (since $\alpha_1 \notin k_1$). By the Tower Law, we have

$$[l_1 : k_1] = [l_1 : k_1(\alpha_1)][k_1(\alpha_1) : k_1] \implies [l_1 : k_1(\alpha_1)] < n.$$

Let m_1 be the minimal polynomial of α_1 over k_1 ($\deg(m_1) = \overline{n}$). Because $f_1(\alpha_1) = 0$, it follows that $m_1|f_1$ in $k_1[x]$.
Define $m_2 = \rho(m_1)$. Then m_2 is a monic irreducible polynomial in $k_2[x]$ and $m_2|f_2$. Because f_2 splits completely in $l_2[x]$, m_2 must also split in $l_2[x]$, so there exists a root $\alpha_2 \in l_2$ such that $m_2(\alpha_2) = 0$.
By the previous lemma, there is an isomorphism $\rho^+ : k_1(\alpha_1) \to k_2(\alpha_2)$ such that $\rho^+|_{k_1} = \rho$ and $\rho^+(\alpha_1) = \alpha_2$.
Notice that $l_1/k_1(\alpha_1)$ is a splitting field extension for f_1, and $l_2/k_2(\alpha_2)$ is a splitting field extension for f_2. Furthermore, the extended isomorphism maps the polynomials correctly: $\rho^+(f_1) = f_2$.

Because the degree of the remaining extension is strictly smaller ($[l_1 : k_1(\alpha_1)] < n$), we can apply the induction hypothesis to ρ^+. By the induction hypothesis, there is an isomorphism $\tau : l_1 \to l_2$ such that $\tau|_{k_1(\alpha_1)} = \rho^+$.
Finally, because τ restricts to ρ^+ on $k_1(\alpha_1)$, and ρ^+ restricts to ρ on k_1, it follows that $\tau|_{k_1} = \rho$. $\square$

Definition 32.4. *Let l/k be a field extension. Define*

$$\mathrm{Aut}(l/k) = \{\sigma \in \mathrm{Aut}(l) : \sigma|_k = 1_k\}.$$

Lemma 32.5. *If $[l : k] < \infty$, then $|\mathrm{Aut}(l/k)| \leq [l : k]$.*

Lemma 32.6. *If $\sigma \in \mathrm{Aut}(l/k)$, $\beta \in l$, and $g \in k[x]$ with $g(\beta) = 0$, then $g(\sigma(\beta)) = 0$.*

Definition 32.7. *A field extension l/k is **algebraic** if every $\alpha \in l$ is algebraic over k.*

33 Lecture 33: Normal Extensions

Theorem 33.1. *Let l/k be a finite field extension (so $[l:k] < \infty$). Let Ω be an algebraically closed field, and let $\phi : k \to \Omega$ be a field monomorphism (embedding). There exists at least one monomorphism $\psi : l \to \Omega$ such that $\psi|_k = \phi$. The total number of such extensions ψ is exactly equal to the separable degree $[l:k]_s$. Consequently, the number of extensions is always less than or equal to $[l:k]$.*

Proof. We will show this by induction on the degree $n = [l:k]$.

Base Case: If $[l:k] = 1$, then $l = k$. The only extension of $\phi : k \to \Omega$ to l is ϕ itself. Thus, there is exactly 1 extension, and $1 \leq 1$. The base case holds.

Inductive Step: Assume the claim holds for all finite extensions of degree less than n, and let $[l:k] = n > 1$.

Choose an element $a \in l \setminus k$. We then have the tower of fields $k \subsetneq k(a) \subseteq l$. Because $a \notin k$, we know $[k(a):k] > 1$. By the Tower Law, $[l:k] = [l:k(a)][k(a):k]$, which implies $[l:k(a)] < [l:k]$.

Claim: There are at most $[k(a):k]$ extensions of ϕ to an embedding $\phi' : k(a) \to \Omega$.

To show this, let $m(x) \in k[x]$ be the minimal polynomial of a over k, so $\deg(m) = [k(a):k]$. Let $\overline{m}(x) \in \Omega[x]$ be the polynomial obtained by applying ϕ to the coefficients of $m(x)$.

For any embedding ϕ' extending ϕ, we have $m(a) = 0$, which implies that $\overline{m}(\phi'(a)) = 0$. Therefore, $\phi'(a)$ must be a root of $\overline{m}(x)$ in Ω. Because Ω is a field, the polynomial $\overline{m}(x)$ can have at most $\deg(\overline{m}) = \deg(m) = [k(a):k]$ distinct roots in Ω.

Since an embedding $\phi' : k(a) \to \Omega$ that extends ϕ is entirely determined by the value of $\phi'(a)$, there are at most $[k(a):k]$ such extensions ϕ'.

Now, by our induction hypothesis, because $[l:k(a)] < [l:k]$, each such embedding $\phi' : k(a) \to \Omega$ can be extended to an embedding $\psi : l \to \Omega$ in at most $[l:k(a)]$ ways.

Multiplying the maximum number of choices at each step, the total number of extensions $\psi : l \to \Omega$ of ϕ is at most:

$$[k(a):k] \times [l:k(a)] = [l:k]$$

This completes the induction step and the proof. □

Theorem 33.2. *If $[l:k] < \infty$ then $|Aut(l/k)| \leq [l:k]$*

Proof. Let $\bar{l}$ be an algebraic closure of l (and thus also an algebraic closure of k).

By definition, the automorphism group $Aut(l/k)$ consists of all field isomorphisms $\sigma : l \to l$ that fix k pointwise (i.e., $\sigma|_k = \mathrm{id}_k$).

Because $l \subseteq \bar{l}$, any such automorphism $\sigma \in Aut(l/k)$ can be viewed as an embedding from l into $\bar{l}$ that extends the identity map $\mathrm{id}_k : k \to \bar{l}$. Therefore, $Aut(l/k)$ is simply a subset of the set of all embeddings $\psi : l \to \bar{l}$ that extend id_k.

By the Extension of Monomorphisms theorem (setting $\Omega = \bar{l}$ and $\phi = \mathrm{id}_k$), the total number of such embeddings ψ is at most $[l : k]$.

Because $Aut(l/k)$ is a subset of these embeddings, it immediately follows that:

$$|Aut(l/k)| \leq [l : k]$$

This completes the proof. □

Definition 33.3. *An extension l/k is normal if*
1. *l/k is algebraic*
2. *For every $\alpha \in l$, minimal polynomial of α/k splits in $l[x]$*

Definition 33.4. *An irreducible $f \in k[x]$ is separable iff f has no repeated roots in any extension.*

Definition 33.5. *A polynomial $g \in k[x]$ is separable iff every irreducible factor is separable.*

Definition 33.6. *Let l/k be algebraic, let $\alpha \in l$ then α is separable iff the minimal polynomial of α/k is separable.*

Definition 33.7. *An extension l/k is separable iff*
1. *l/k*
2. *Every $\alpha \in l$ separable*

Theorem 33.8. *TFAE for l/k, $[l : k] < \infty$:*
1. *l/k normal*
2. *l/k sfe for some $f \in k[x]$*

Proof. To show (1) $\implies$ (2), find $a_1, ..., a_n \in l$ such that $l = k(a_1, a_2, ..., a_n)$, let $f = \prod_{i=1}^{n} m_{a_i}$ where m_{a_i} is the minimal polynomial of a_i/k.
To show (2) $\implies$ (1), let $\alpha \in l$ we need to show that if m is the minimal polynomial of α/k then m splits in $l[x]$. Let l' be the sfe of l for m. To show $l' = l$, it suffices to show that if $\beta \in l'$ and β a root of m then $\beta \in l$.
m monic irreducible in $k[x]$, $m(\beta) = 0$ so m is minimal polynomial of β/k. So there is an IM $\psi : k(\beta) \to k(\beta)$ with $\psi(\alpha) = \beta$, $\psi|_k = 1_k$. We have $[k(\alpha) : k] = [k(\beta) : k]$.
The key points are

1. l/k sfe for f so $l(\beta)/k(\beta)$ sfe for f
2. l/k sfe for f so $l/k(\alpha)$ sfe for f
3. $\psi(f) = f$
4. ψ IM

So ψ lifts to an isomorphism $\rho : l \xrightarrow{\sim} l(\beta)$ with $\rho|_{k(\alpha)} = \psi$. Using ρ, we have $[l : k(\alpha)] = [l(\beta) : k(\beta)]$. Since $[l : k]$ is finite and $[k(\alpha) : k] = [k(\beta) : k]$, we conclude

$[l(\beta) : k] = [l : k]$, forcing $l(\beta) = l$, i.e. $\beta \in l$. Hence every root of m lies in l, so m splits over l. $\square$

34 Lecture 34: Field Extensions

Definition 34.1. *k' is an **intermediate field** of l/k if $k \subseteq k' \subseteq l$.*

Lemma 34.2. *If $k \subseteq k' \subseteq l$, then $\mathrm{Aut}(l/k') \leq \mathrm{Aut}(l/k) \leq \mathrm{Aut}(l)$.*

Definition 34.3. *If $H \leq \mathrm{Aut}(l)$ and l is a field, then*

$$\mathrm{Fix}(H) = \{a \in l : \sigma(a) = a \text{ for all } \sigma \in H\}.$$

Lemma 34.4. *The following are true:*
1. *$\mathrm{Fix}(H)$ is a subfield of l.*
2. *If $H \leq \mathrm{Aut}(l/k)$, then $\mathrm{Fix}(H)$ is an intermediate field of l/k.*
3. *$k \leq k' \leq l \implies \mathrm{Aut}(l/k') \leq \mathrm{Aut}(l/k)$.*

Definition 34.5. *Let $f \in k[x]$ with*

$$f = \sum_{i=0}^{n} c_i x^i. \qquad (c_i \in k)$$

The formal derivative of f is

$$f' = \sum_{i=1}^{n} i c_i x^{i-1}.$$

Lemma 34.6. *Let $f \in k[x]$ with $a \in k$ then*

$$f = (x-a)^2 q + \lambda(x-a) + \mu.$$

We have

$$f(a) = \mu, f'(a) = \lambda.$$

So $(x-a)^2 | f \iff f(a) = f'(a) = 0$

Lemma 34.7. *Let $f \in k[x]$ be irreducible. If $f' \neq 0$, then $\gcd(f, f') = 1$.*
Since $k[x]$ is a PID, there exist $A, B \in k[x]$ such that $Af + Bf' = 1$. If l/k is any extension and $a \in l$, this equation is still valid in $l[x]$, giving $A(a)f(a) + B(a)f'(a) = 1$. So we cannot have $f(a) = f'(a) = 0$, meaning f has no repeated roots in l.

Theorem 34.8. *If irreducible $f \in k[x]$ is inseparable then $f' = 0$.*

Theorem 34.9. *If $f \in k[x]$ separable and l/k then f still separable in $l[x]$.*

Proof. Suppose not; let $g \in l[x]$ be an inseparable irreducible factor of f. Then g has a repeated root in some extension l'/l. Factor $f = f_1 \cdots f_t$ in $k[x]$ with each f_i irreducible. Then $g \mid f_1 \cdots f_t$ in $l[x]$, so $g \mid f_i$ for some i (since g is prime). So f_i has a repeated root in l', contradicting the separability of f in $k[x]$. $\square$

Theorem 34.10. *Let l/k be a finite field extension (so $[l:k] < \infty$). The following are equivalent:*

1. $|\mathrm{Aut}(l/k)| = [l:k]$.
2. *l/k is normal and separable.*
3. *l/k is a splitting field extension for some separable $f \in k[x]$.*
4. $k = \mathrm{Fix}(\mathrm{Aut}(l/k))$.

Proof. (2) $\implies$ (3): Write $l = k(a_1, \ldots, a_t)$ for some $a_i \in l$ (possible since $[l:k] < \infty$). Let $f = \prod_{i=1}^{t} m_{a_i}$, where m_{a_i} is the minimal polynomial of a_i over k. Separability of l/k means each m_{a_i} is separable, so f is separable. Normality of l/k means each m_{a_i} splits in $l[x]$, so f splits in $l[x]$. Since $l = k(a_1, \ldots, a_t)$ and each a_i is a root of f, l is generated over k by roots of f; hence l/k is the sfe for f.

(3) $\implies$ (1): By induction on $[l:k]$.
(Base Case) $[l:k] = 1$ means $l = k$, so $\mathrm{Aut}(l/k) = \{1\}$ and $|\mathrm{Aut}(l/k)| = 1 = [l:k]$.
(Inductive Step) If $l \neq k$, let $a \in l \setminus k$ be a root of f (possible since l/k is generated by such roots), and let $m = m_a$ be the minimal polynomial of a over k. Since $f(a) = 0$, $m \mid f$ in $k[x]$. Since l/k is sfe for f, it is normal, so m splits in $l[x]$; since f is separable, so is m, and m splits as $\prod_{i=1}^{t}(x - a_i)$ with distinct roots $a_i \in l$. Set $a_1 = a$; for each i, $m_{a_i} = m$. So for each $i \in \{1, \ldots, t\}$ there is a unique isomorphism

$$\pi_i : k(a) \to k(a_i) \text{ with } \pi_i|_k = 1_k \text{ and } \pi_i(a) = a_i.$$

Now l is the sfe of $k(a)$ for f and also of $k(a_i)$ for f, and π_i fixes the coefficients of f (which lie in k). By the uniqueness of splitting field extensions, π_i extends to an isomorphism $\sigma_i : l \to l$, and $\sigma_i \in \mathrm{Aut}(l/k)$ with $\sigma_i(a) = a_i$.

Each element of $\mathrm{Aut}(l/k)$ restricts to some π_i on $k(a)$, and by induction (since $[l:k(a)] < [l:k]$), there are exactly $[l:k(a)]$ extensions of each π_i. Thus $|\mathrm{Aut}(l/k)| = t \cdot [l:k(a)] = [k(a):k] \cdot [l:k(a)] = [l:k]$.

(1) $\implies$ (2): Suppose for contradiction that l/k is not normal. Let $a \in l$ be such that m_a fails to split in $l[x]$. Then the number of roots of m_a in l is less than $\deg(m_a) = [k(a):k]$. Enumerate the roots of m_a in l as $a_1, \ldots, a_t$ with $a_1 = a$, where $t < [k(a):k]$. If $\sigma \in \mathrm{Aut}(l/k)$, then $m_a(a) = 0 \implies m_a(\sigma(a)) = 0$, so $\sigma(a) = a_i$ for some $1 \leq i \leq t$. So $\sigma|_{k(a)} = \pi_i$ for a unique isomorphism $\pi_i : k(a) \simeq k(a_i)$ extending 1_k. Each σ extending π_i is an extension of an embedding $k(a) \hookrightarrow l$; by the extension theorem there are at most $[l:k(a)]$ such extensions. Therefore

$$|\mathrm{Aut}(l/k)| \leq t \cdot [l:k(a)] < [k(a):k] \cdot [l:k(a)] = [l:k],$$

contradicting (1).
If l/k is normal but not separable, choose $a \in l$ with m_a inseparable. Since m_a splits in l with at most $\deg(m_a)$ distinct roots but has fewer (due to repeated roots), the same counting argument gives $|\mathrm{Aut}(l/k)| < [l:k]$, contradiction.

(1) $\implies$ (4): We prove a stronger statement (Artin's Theorem): Let l be a field, $G \leq \mathrm{Aut}(l)$ finite with $|G| = n$, and let $k = \mathrm{Fix}(G)$. Then

(i) $[l:k] = n$.

(ii) $G = \mathrm{Aut}(l/k)$.

(iii) l/k is Galois.

Since $G \leq \mathrm{Aut}(l/k)$, we have $|\mathrm{Aut}(l/k)| \geq n$, and combined with the bound $|\mathrm{Aut}(l/k)| \leq [l : k]$, it cannot be that $[l : k] < n$.
Suppose for contradiction that $[l : k] > n$. Choose $a_1, \ldots, a_{n+1} \in l$ linearly independent over k. Enumerate $G = \{\rho_1, \ldots, \rho_n\}$ with $\rho_1 = 1_l$. For $j = 1, \ldots, n+1$, set

$$v_j = \begin{pmatrix} \rho_1(a_j) \\ \vdots \\ \rho_n(a_j) \end{pmatrix} \in l^n.$$

The space l^n has dimension n over l, so the $n+1$ vectors $v_1, \ldots, v_{n+1}$ are linearly dependent over l. Let s be minimal such that some subset of size s is dependent; after relabeling, $v_1, \ldots, v_s$ are dependent. So there exist $\lambda_1, \ldots, \lambda_s \in l$, not all zero, with

$$\sum_{i=1}^{s} \lambda_i v_i = 0.$$

By minimality of s, all $\lambda_i \neq 0$. Scaling, we may assume $\lambda_1 = 1$.
Looking at the first component of the vector equation,

$$\sum_{j=1}^{s} \lambda_j \rho_1(a_j) = \sum_{j=1}^{s} \lambda_j a_j = 0. \tag{$*$}$$

If all λ_j lay in $k = \mathrm{Fix}(G)$, this would contradict k-linear independence of the a_j. So some $\lambda_j \notin k$; relabel so $\lambda_s \notin k$, and pick $\rho_n \in G$ with $\rho_n(\lambda_s) \neq \lambda_s$.
Apply ρ_n to each component of the equation $\sum_j \lambda_j v_j = 0$: component i gives $\sum_j \rho_n(\lambda_j)\rho_n(\rho_i(a_j)) = 0$. Since $\rho_n G = G$, as i ranges over $1, \ldots, n$, the map $\rho_n \rho_i$ also ranges over all of G. Reindexing, we get

$$\sum_{j=1}^{s} \rho_n(\lambda_j)\rho_i(a_j) = 0 \quad \text{for all } i = 1, \ldots, n,$$

i.e.

$$\sum_{j=1}^{s} \rho_n(\lambda_j) v_j = 0.$$

Subtracting from $\sum_j \lambda_j v_j = 0$,

$$\sum_{j=1}^{s} (\lambda_j - \rho_n(\lambda_j)) v_j = 0.$$

Since $\lambda_1 - \rho_n(\lambda_1) = 1 - 1 = 0$, this is a nontrivial dependence among $v_2, \ldots, v_s$ (nontrivial because the coefficient of v_s is $\lambda_s - \rho_n(\lambda_s) \neq 0$), contradicting minimality of s. Therefore $[l : k] = n$.
Since $G \leq \mathrm{Aut}(l/k)$ with $|G| = n$ and $|\mathrm{Aut}(l/k)| \leq [l : k] = n$, we get $G = \mathrm{Aut}(l/k)$. Finally $|\mathrm{Aut}(l/k)| = n = [l : k]$, so l/k satisfies (1), hence is Galois.
To finish the equivalences in the original theorem: applying the stronger statement with $G = \mathrm{Aut}(l/k)$ and $k' = \mathrm{Fix}(G)$, we get $[l : k'] = |G|$ and $G = \mathrm{Aut}(l/k')$. Then

$$(1) \iff |G| = [l : k] \iff [l : k'] = [l : k] \iff k' = k \iff (4).$$

$\square$

Definition 34.11. *A finite extension l/k is* ***Galois*** *if it satisfies any of the equivalent conditions above.*

Theorem 34.12 (Fundamental Theorem of Galois Theory)**.** *Let l/k be a finite Galois extension. Then:*

1. *l/k' is Galois for every intermediate field k'.*
2. *The operations*
$$k' \mapsto \mathrm{Aut}(l/k') \quad \textit{and} \quad H \mapsto \mathrm{Fix}(H)$$
give mutually inverse, inclusion-reversing bijections between
   - *intermediate fields k' with $k \subseteq k' \subseteq l$, and*
   - *subgroups $H \leq \mathrm{Aut}(l/k)$.*
3. *For an intermediate field k', k'/k is Galois iff $\mathrm{Aut}(l/k') \trianglelefteq \mathrm{Aut}(l/k)$; in this case*
$$\mathrm{Aut}(k'/k) \simeq \frac{\mathrm{Aut}(l/k)}{\mathrm{Aut}(l/k')}.$$

Proof. We address each part:

1. Since l/k is an sfe for some separable $f \in k[x]$, and f remains separable when viewed in $k'[x]$, l/k' is an sfe for f and hence Galois.
2. If k' is intermediate, then since l/k' is Galois, $k' = \mathrm{Fix}(\mathrm{Aut}(l/k'))$. Conversely, if $H \leq \mathrm{Aut}(l/k)$, then H is finite, so by Artin's theorem $H = \mathrm{Aut}(l/\mathrm{Fix}(H))$.
3. Proven in the next lecture (see Lecture 35).

□

35 Lecture 35: Radical

We complete the proof of the Fundamental Theorem of Galois Theory, Part (3): L/K Galois, $[L : K] < \infty$, K' intermediate. Then K'/K is Galois $\iff$ $\mathrm{Aut}(L/K') \trianglelefteq \mathrm{Aut}(L/K)$, and in this case

$$\mathrm{Aut}(K'/K) \cong \frac{\mathrm{Aut}(L/K)}{\mathrm{Aut}(L/K')}.$$

Proof. **($\Rightarrow$):** Assume K'/K Galois. Let $\sigma \in \mathrm{Aut}(L/K)$, $a \in K'$ such that m_a splits in $K'[x]$:

$$m_a := \prod_{i=1}^{t}(x - a_i), \quad a_i \in K'.$$

For all $\sigma \in \mathrm{Aut}(L/K)$, $m(a) = 0 \implies m(\sigma(a)) = 0$, so $\sigma(a) = a_i$ for some i. Therefore $\sigma|_{K'} \in \mathrm{Aut}(K'/K)$, and $\sigma \mapsto \sigma|_{K'}$ is a homomorphism

$$\mathrm{Aut}(L/K) \to \mathrm{Aut}(K'/K), \quad \text{kernel} = \mathrm{Aut}(L/K').$$

Since L/K and L/K' are both Galois,

$$|\mathrm{Aut}(L/K)| = [L : K], \quad |\mathrm{Aut}(L/K')| = [L : K'].$$

So

$$\frac{|\mathrm{Aut}(L/K)|}{|\mathrm{Aut}(L/K')|} = [K' : K].$$

Thus $|\mathrm{im}(\text{restriction})| = [K' : K]$ and $\mathrm{im} = \mathrm{Aut}(K'/K)$. By the first isomorphism theorem,

$$\mathrm{Aut}(K'/K) \cong \frac{\mathrm{Aut}(L/K)}{\mathrm{Aut}(L/K')}.$$

In particular $\mathrm{Aut}(L/K') \trianglelefteq \mathrm{Aut}(L/K)$.

($\Leftarrow$): We saw $\mathrm{Aut}(L/K') \trianglelefteq \mathrm{Aut}(L/K)$ if K'/K Galois. Now suppose $\mathrm{Aut}(L/K') \trianglelefteq \mathrm{Aut}(L/K)$; we want to show K'/K is Galois.

Since L/K is separable, K'/K is separable for every intermediate K'. It remains to show K'/K is normal.

Let $a \in K'$, $m = m_a^K$ the minimal polynomial. Since L/K is normal, m splits in $L[x]$:

$$m = \prod(x - a_i).$$

By HWII, for all i there exists $\sigma \in \mathrm{Aut}(L/K)$ such that $\sigma(a) = a_i$.

Claim 35.1. *$a_i \in K'$ for all i.*

Proof. Since L/K' is Galois, $K' = Fix(\mathrm{Aut}(L/K'))$. So it suffices to show $a_i \in Fix(\mathrm{Aut}(L/K'))$.

Fix $\sigma \in \mathrm{Aut}(L/K)$ with $\sigma(a) = a_i$. Fix $\tau \in \mathrm{Aut}(L/K')$. Since $\mathrm{Aut}(L/K') \trianglelefteq \mathrm{Aut}(L/K)$, we have $\sigma^{-1}\tau\sigma \in \mathrm{Aut}(L/K')$. Since $a \in K'$:

$$(\sigma^{-1}\tau\sigma)(a) = a.$$

Therefore $\tau\sigma(a) = \sigma(a)$, i.e., $\tau(a_i) = a_i$. $\square$

Since all roots a_i of m_a^K lie in K', m splits in $K'[x]$, so K'/K is normal. Combined with separability, K'/K is Galois. $\square$

Definition 35.2 (Radical Extension)**.** *A field extension L/K is **radical** if and only if there exist intermediate fields*

$$K = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_s = L$$

such that for all i, $K_{i+1} = K_i(\alpha_i)$ where there exists $n_i > 0$ with $\alpha_i^{n_i} \in K_i$.

Definition 35.3 (Solvable by Radicals)**.** *A polynomial $f \in K[x]$ is **solvable by radicals** if and only if there exists a radical extension L/K with an intermediate field that is a splitting field extension of K for f.*

Until further notice, all fields have characteristic 0.

Theorem 35.4. *The following are equivalent:*
1. *f is solvable by radicals.*
2. *If L/K is the splitting field extension over K for f, then $\mathrm{Aut}(L/K)$ is solvable.*

Theorem 35.5. *For any field K (any characteristic), any finite subgroup of K^* (under multiplication) is cyclic.*

Definition 35.6 (Primitive Root of Unity)**.** *For any field K (any characteristic), a **primitive n-th root of unity** is an element $a \in K^*$ that is a generator of $\{a : a^n = 1\}$, i.e., a has order n in K^*. Equivalently, $\mathrm{ord}(a) = n$ in K^*.*

Example 35.7. *In $\mathbb{C}$, $e^{2\pi i/n}$ is a primitive n-th root of unity.*

Let K have characteristic zero. Consider $f(x) := x^n - 1 \in K[x]$. Then $f'(x) = nx^{n-1} \neq 0$ since $\mathrm{char}(K) = 0$, so f has no repeated roots in any extension L/K, hence f is separable.

Let L/K be the splitting field extension for $x^n - 1$. Since f splits without repeated factors, there are exactly n distinct n-th roots of 1 in L. The set $\{a \in L^* : a^n = 1\} \leq L^*$ is a finite subgroup, hence cyclic of order n, generated by some primitive n-th root of unity α. Moreover $L = K(\alpha)$.

Let $\sigma \in \mathrm{Aut}(L/K)$. Then σ is determined by $\sigma(\alpha)$, which must also be a primitive n-th root, so $\sigma(\alpha) = \alpha^i$ for some i. It follows that $\mathrm{Aut}(L/K)$ is abelian: for any $\sigma, \tau \in \mathrm{Aut}(L/K)$, $\sigma \circ \tau = \tau \circ \sigma$ since both map α to the same power.

More generally, consider $g(x) := x^n - a$ where $a \in K$, $a \neq 0$. We have $g'(x) = nx^{n-1} \neq 0$, so g has no repeated roots. Let L/K be the SFE for g. In $L[x]$,

$$x^n - a = \prod_{i=1}^{n} (x - \alpha_i), \quad \alpha_i^n = a, \quad \alpha_i \text{ distinct.}$$

Consider $\frac{\alpha_i}{\alpha_1}$: we have $\left(\frac{\alpha_i}{\alpha_1}\right)^n = \frac{a}{a} = 1$. So $x^n - 1$ splits in L, giving n distinct n-th roots

of unity. If ω is a primitive n-th root of unity and $\alpha^n = a$, then

$$x^n - a = \prod_{i=0}^{n-1}(x - \omega^i \alpha).$$

So $L = K(\omega^i \alpha \mid i < n) = K(\omega, \alpha)$.

In all cases, $K \subseteq K(\omega) \subseteq K(\omega, \alpha)$, where $\mathrm{Aut}(K(\omega)/K)$ is abelian and $\mathrm{Aut}(K(\omega, \alpha)/K(\omega))$ is abelian. Therefore $\mathrm{Aut}(L/K)$ is solvable.

36 Lecture 36

Theorem 36.1. *Let $a \in k$ with k field and l/k sfe for $x^n - a$ then $x^n - a = \prod_{i=1}^{n}(x - \alpha_i)$ with $\alpha_i \in l$ distinct.*

Proof. Let α_1 be any root of $x^n - a$ in l. For any other root α_i, we have $(\alpha_i/\alpha_1)^n = a/a = 1$, so α_i/α_1 is an n-th root of unity in l. Hence there is a primitive n-th root of unity $\zeta \in l$, and we can enumerate the roots as $\alpha_i = \alpha\zeta^i$ for $0 \leq i < n$, where α is one chosen root with $\alpha^n = a$. Thus $l = k(\alpha, \zeta)$, and l/k is Galois (the sfe of a separable polynomial).

Claim 36.2. $\mathrm{Aut}(l/k)$ *is solvable.*

Proof. Consider the tower $k \subseteq k(\zeta) \subseteq k(\alpha, \zeta) = l$.

- $k(\zeta)/k$ is Galois (it is the sfe for $x^n - 1 = \prod_{i=0}^{n-1}(x - \zeta^i)$). For any $\sigma \in \mathrm{Aut}(k(\zeta)/k)$, $\sigma(\zeta) = \zeta^j$ for some j coprime to n, so $\mathrm{Aut}(k(\zeta)/k)$ embeds in $(\mathbb{Z}/n\mathbb{Z})^*$ and is abelian.
- Consider $\mathrm{Aut}(k(\alpha, \zeta)/k(\zeta))$. If σ is in this group, then $\sigma(\alpha)^n = \sigma(\alpha^n) = \sigma(a) = a$, so $\sigma(\alpha) = \alpha\zeta^i$ for some i. Thus σ is determined by i, and $\mathrm{Aut}(k(\alpha, \zeta)/k(\zeta))$ embeds in $\mathbb{Z}/n\mathbb{Z}$, hence is abelian.

Since $k(\zeta)/k$ is Galois, by the Fundamental Theorem of Galois Theory $\mathrm{Aut}(l/k(\zeta)) \trianglelefteq \mathrm{Aut}(l/k)$ and

$$\frac{\mathrm{Aut}(l/k)}{\mathrm{Aut}(l/k(\zeta))} \simeq \mathrm{Aut}(k(\zeta)/k).$$

Since both $\mathrm{Aut}(l/k(\zeta))$ and $\mathrm{Aut}(k(\zeta)/k)$ are abelian, $\mathrm{Aut}(l/k)$ is solvable. □

□

Example 36.3. *Consider $f = \prod_{i=1}^{t}(x^n - b_i)$ with $b_i \in k$, and let l/k be the sfe of k for f. Then*

1. *$l = k(\zeta, \beta_1, \ldots, \beta_t)$ with ζ a primitive n-th root of 1 and $\beta_i^n = b_i$.*
2. *$\mathrm{Aut}(l/k)$ is solvable.*

Proof. **Goal:** If $f \in k[x]$ is solvable by radicals and l/k is the sfe for f, then $\mathrm{Aut}(l/k)$ is solvable.
Subgoal: If l_2/k is a radical extension, then there exists $l_3 \supseteq l_2$ such that

1. l_3/k is Galois,
2. $\mathrm{Aut}(l_3/k)$ is solvable,
3. l_3/k is radical.

Subgoal $\implies$ Goal. By hypothesis, we have $k \subseteq l_1 \subseteq l_2$ with l_1/k the sfe for f and l_2/k radical. By the subgoal, extend to $k \subseteq l_1 \subseteq l_2 \subseteq l_3$ with l_3/k Galois and $\mathrm{Aut}(l_3/k)$ solvable. Since l_3/k is Galois with intermediate field l_1 and l_1/k is Galois (being an sfe of a separable polynomial), by the Fundamental Theorem $\mathrm{Aut}(l_1/k) \simeq \mathrm{Aut}(l_3/k)/\mathrm{Aut}(l_3/l_1)$. Quotients of solvable groups are solvable, so $\mathrm{Aut}(l_1/k)$ is solvable.

Proof of Subgoal. We proceed by induction on the number of steps in the radical

tower l_2/k.
(Base Case) $l_2 = k(\alpha)$ where $\alpha^n = a \in k$. Let l_3/l_2 be the sfe for $x^n - 1$, so $l_3 = l_2(\zeta)$ with ζ a primitive n-th root of 1. Then $l_3 = k(\alpha, \zeta)$ is the sfe over k of $x^n - a$, hence l_3/k is Galois, is radical (we have $k \subseteq k(\zeta) \subseteq k(\zeta, \alpha)$), and $\mathrm{Aut}(l_3/k)$ is solvable by the previous argument.
(Inductive Step) Suppose $l_2 = l_2^{(0)}(\beta)$ where $\beta^n = b \in l_2^{(0)}$ and $l_2^{(0)}/k$ is a radical extension satisfying the IH: there is $l_3^{(0)} \supseteq l_2^{(0)}$ with $l_3^{(0)}/k$ Galois, $\mathrm{Aut}(l_3^{(0)}/k)$ solvable, and $l_3^{(0)}/k$ radical. Fix $h \in k[x]$ such that $l_3^{(0)}/k$ is the sfe for h. Let $b_1 = b, b_2, \ldots, b_t$ be the orbit of b under $\mathrm{Aut}(l_3^{(0)}/k)$, and set

$$f = \prod_{i=1}^{t} (x^n - b_i).$$

Every $\sigma \in \mathrm{Aut}(l_3^{(0)}/k)$ permutes the factors $x^n - b_i$, so the coefficients of f are $\mathrm{Aut}(l_3^{(0)}/k)$-invariant, i.e. $f \in k[x]$. Let l_3 be the sfe of $l_3^{(0)}$ for f; equivalently, l_3 is the sfe of k for hf. Then l_3/k is Galois and contains l_2. The extension $l_3/l_3^{(0)}$ is a tower of extensions by n-th roots, so it is radical and $\mathrm{Aut}(l_3/l_3^{(0)})$ is solvable. By the Fundamental Theorem of Galois Theory,

$$\mathrm{Aut}(l_3^{(0)}/k) \simeq \mathrm{Aut}(l_3/k)/\mathrm{Aut}(l_3/l_3^{(0)}),$$

so $\mathrm{Aut}(l_3/k)$ is an extension of a solvable group by a solvable group, hence solvable. Also l_3/k is radical (by concatenating the radical towers for $l_3^{(0)}/k$ and $l_3/l_3^{(0)}$). $\square$